



Data Processing

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AI Breakthroughs for Public



If you have an interest in the principle of AlphaGo, please access to <https://www.nature.com/articles/nature16961>

AI Breakthroughs for Public

GatesNotes THE BLOG
OF BILL GATES

LOG IN

SIGN UP



A NEW ERA

The Age of AI has begun

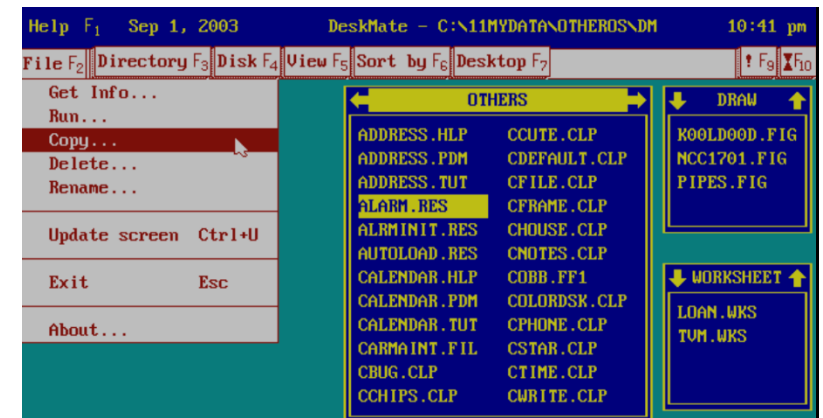
Artificial intelligence is as revolutionary as mobile phones and the Internet.

By **Bill Gates** | March 21, 2023 • 14 minute read



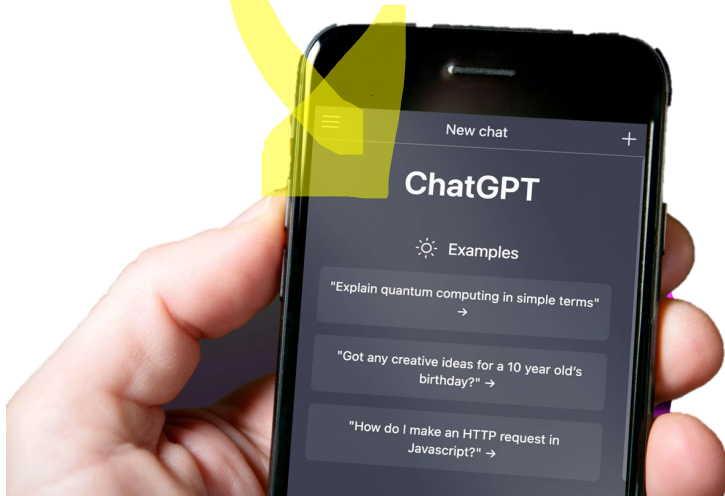
In my lifetime, I've seen **two** demonstrations of technology that struck me as **revolutionary**.

The first time was in 1980, when I was introduced to a **graphical user interface**



AI Breakthroughs for Public

The second big surprise came just last year. I'd been meeting with the team from OpenAI since 2016 and was impressed by their steady progress. In mid-2022, I was so excited about their work that **I gave them a challenge**: train an artificial intelligence **to pass an Advanced Placement biology exam**. Make it capable of answering questions that it hasn't been specifically trained for.



2~3년은
걸리겠지?ㅋㅋㅋ

AI Breakthroughs for Public

In September, when I met with them again, I watched in awe as they asked GPT, their AI model, 60 multiple-choice questions from the AP Bio exam—and it got 59 of them right.

"생물학 시험 60개중 59문제 맞춘 AI에 경외감 느껴"

게이츠는 "지난해 9월 다시 만났을 때 AI가 AP 생물학 시험 60문제를 풀고 59문제를 맞혔다" "나는 그것을 경외감으로 지켜봤다"라고 했다.



말도
안돼!!

AI Breakthroughs for Public



The interactive dialogue interface used to talk with the AI

The ability to create new, original text and content rather than just fetching static data

Team Project: Image Generation

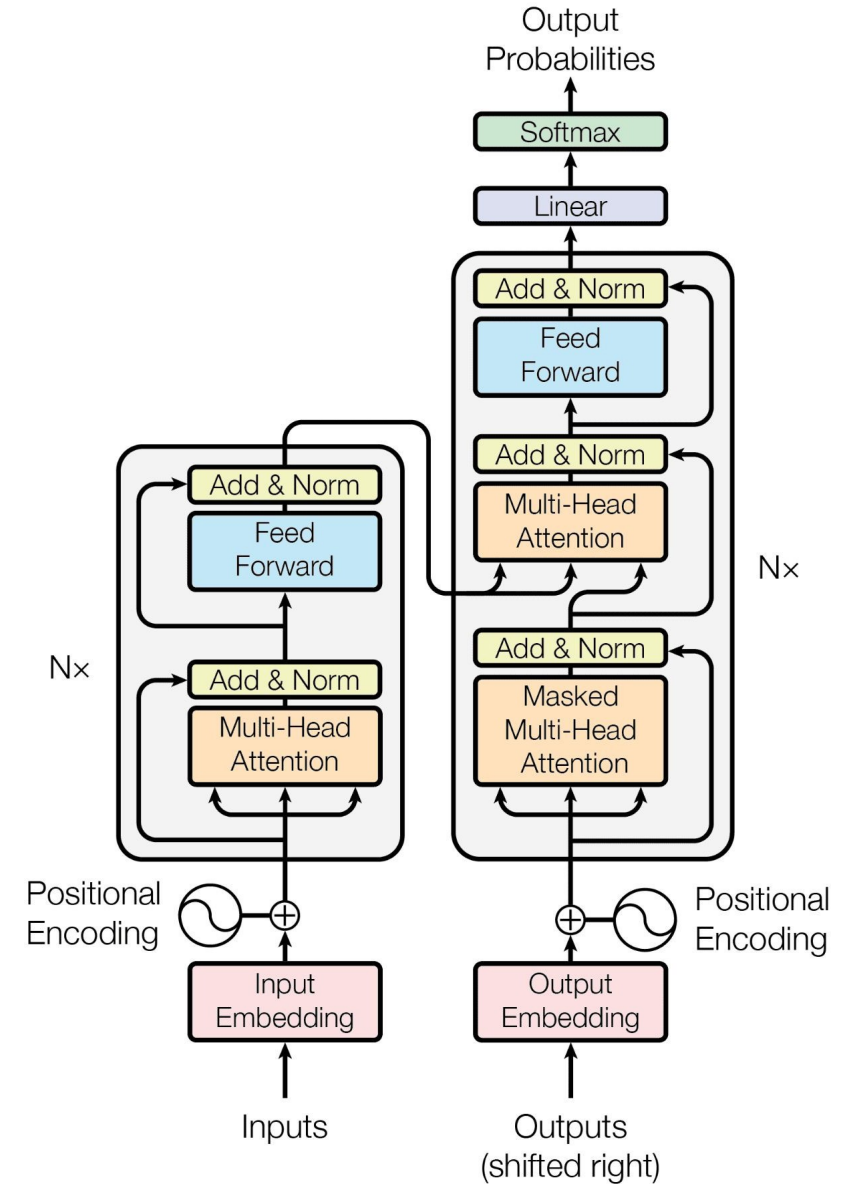
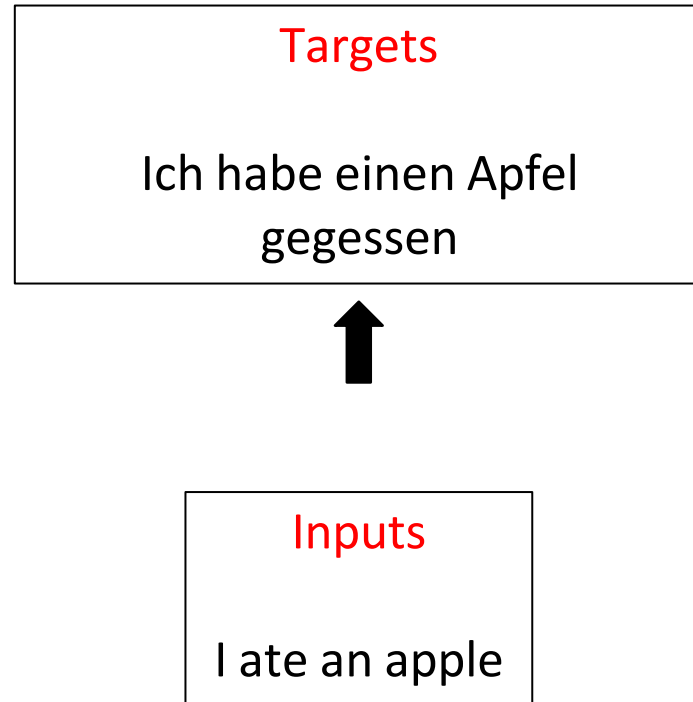
The initial training phase using vast amounts of internet data to learn human language patterns

Team Project: SSL Pretrained Model at Medical Imaging

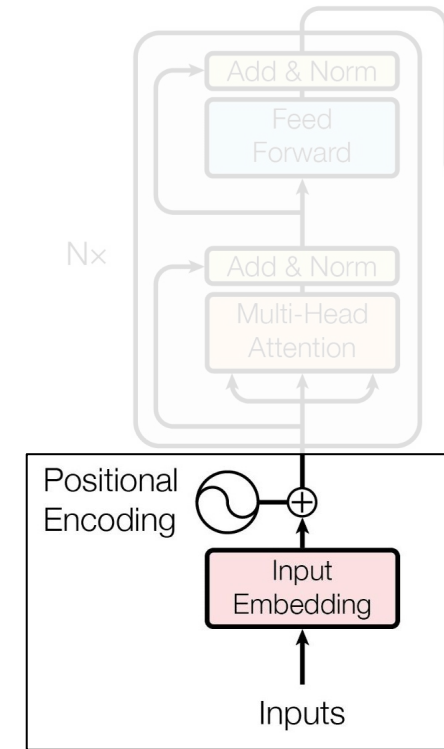
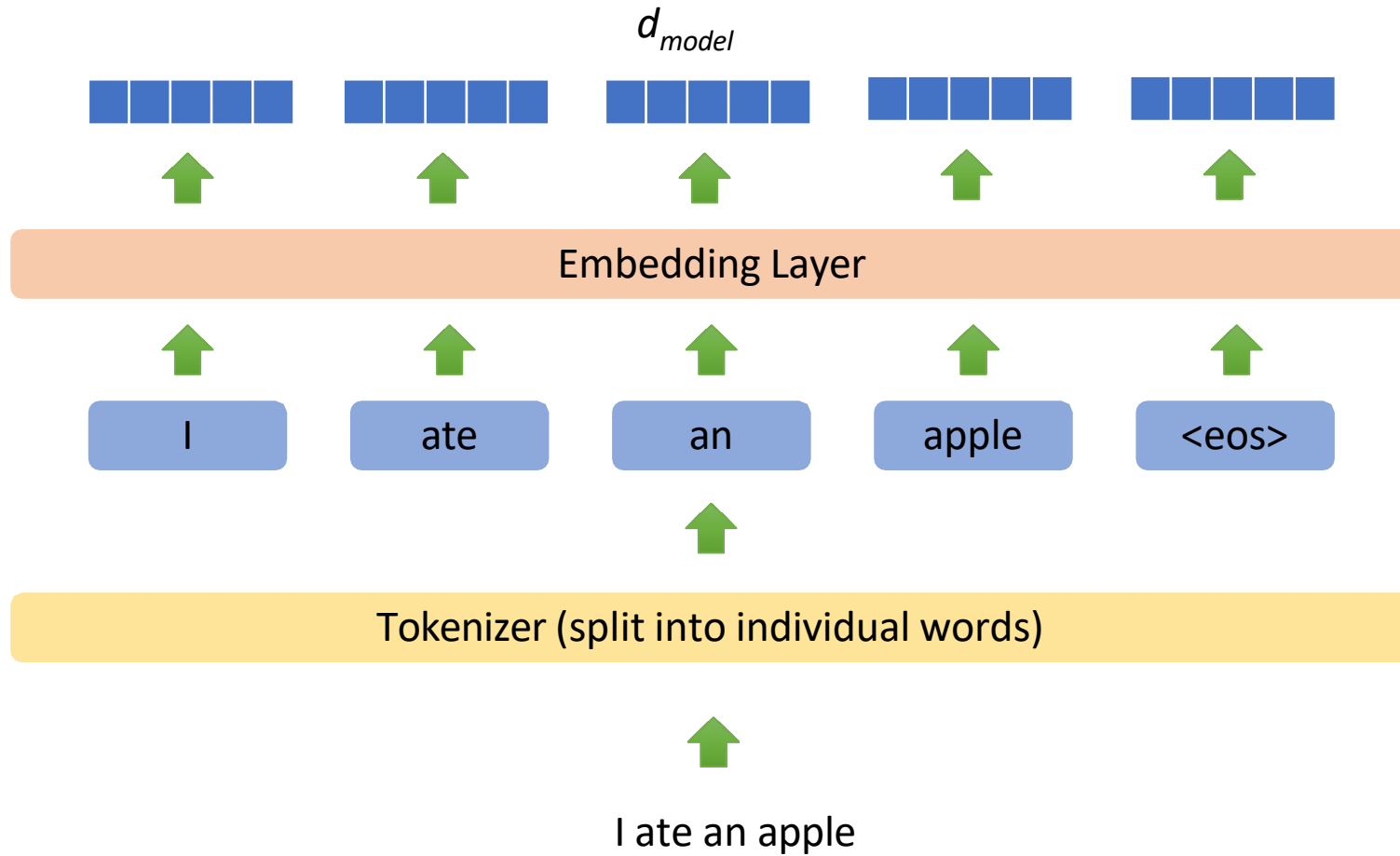
The core neural network architecture designed to process language and understand context

Team Project: NLP Tasks

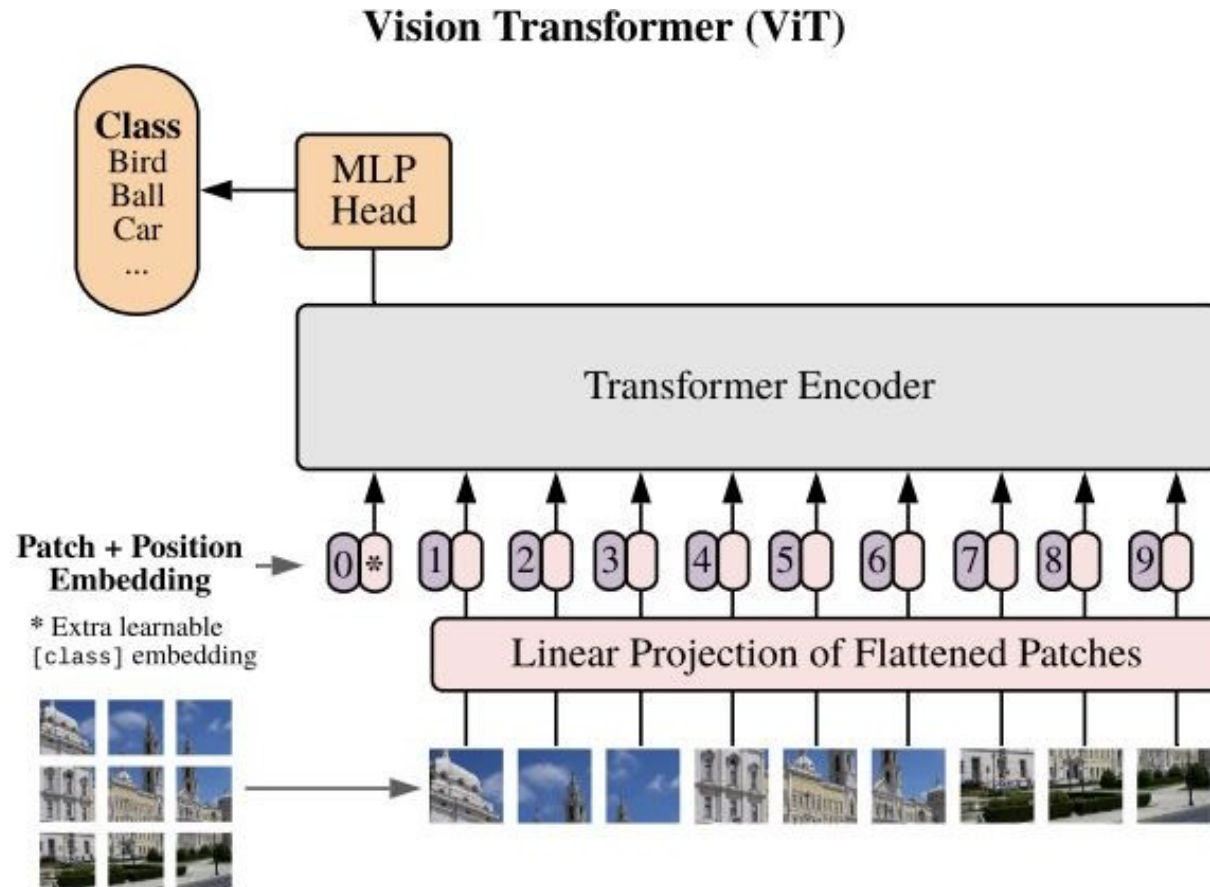
Era of Transformer



Era of Transformer

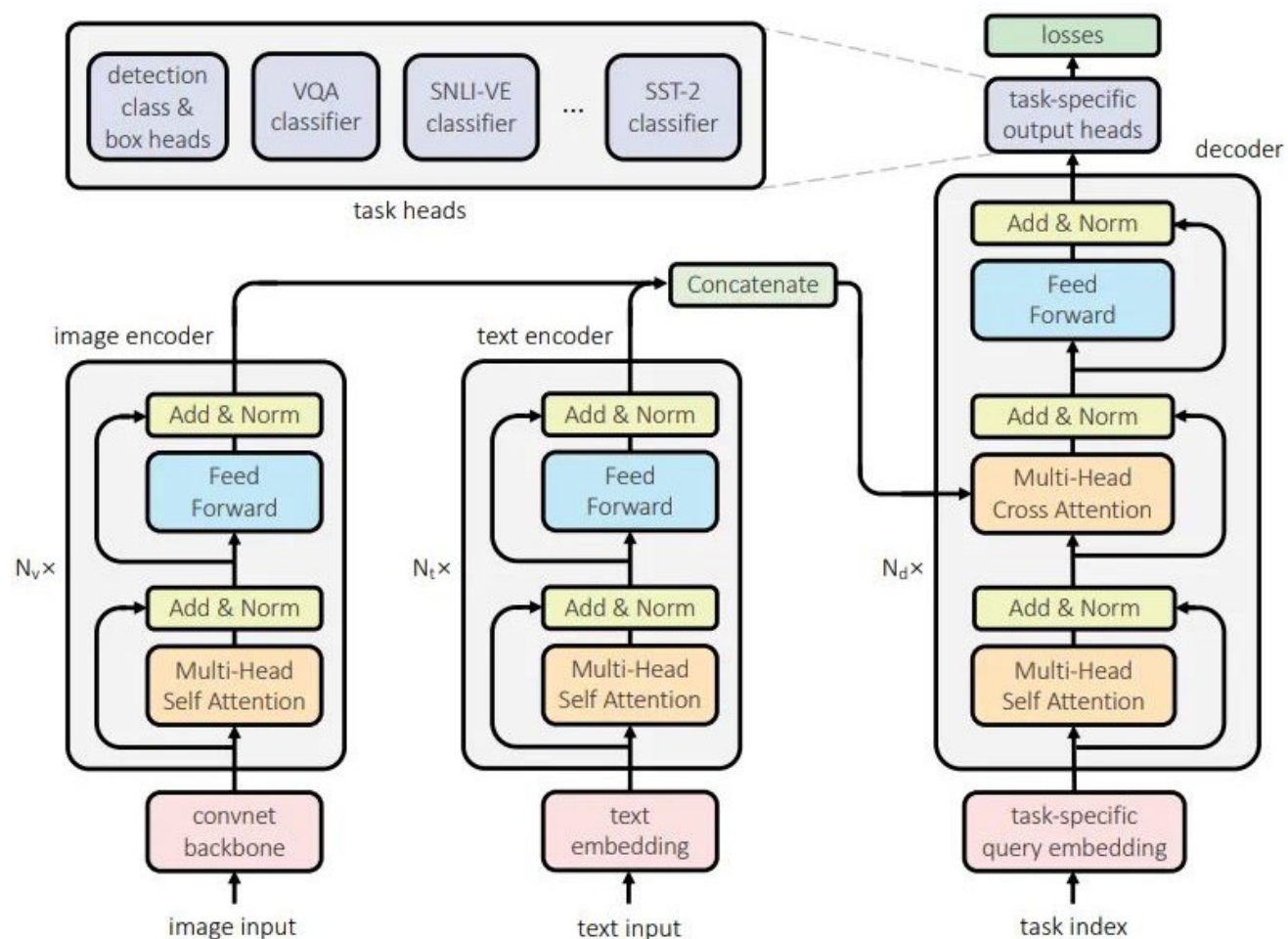


Era of Transformer



1. Split an image into fixed-size patches (16x16 pixels).
2. Tokenize each patch (linear projection of flattened patches).
3. Add position embedding.
4. Feed the resulting sequence of vectors to a standard Transformer encoder.
5. For classification, add an extra learnable “classification token” to the sequence.

Era of Transformer



1. UniT handles 7 tasks ranging from object detection to vision-and language reasoning and natural language understanding.
2. Components:
 - An image encoder to encode the visual inputs.
 - A text encoder to encode the language inputs.
 - A joint decoder with per-task query embedding.
 - Task-specific heads to make the final outputs for each task.

Era of Transformer

Start a new conversation, and the history is cleared.



User

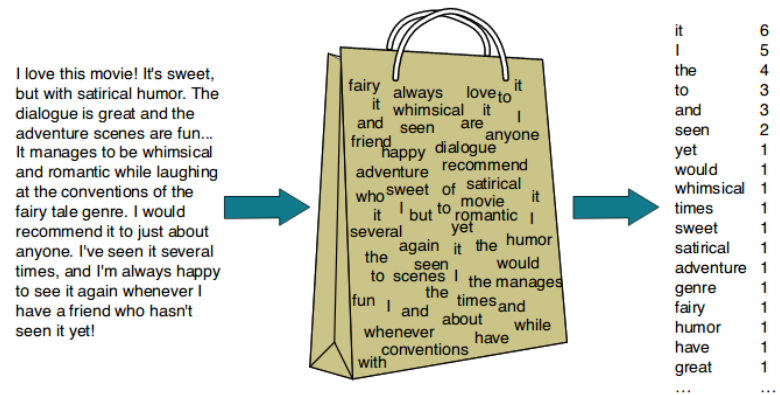
Do you know who drew this painting?



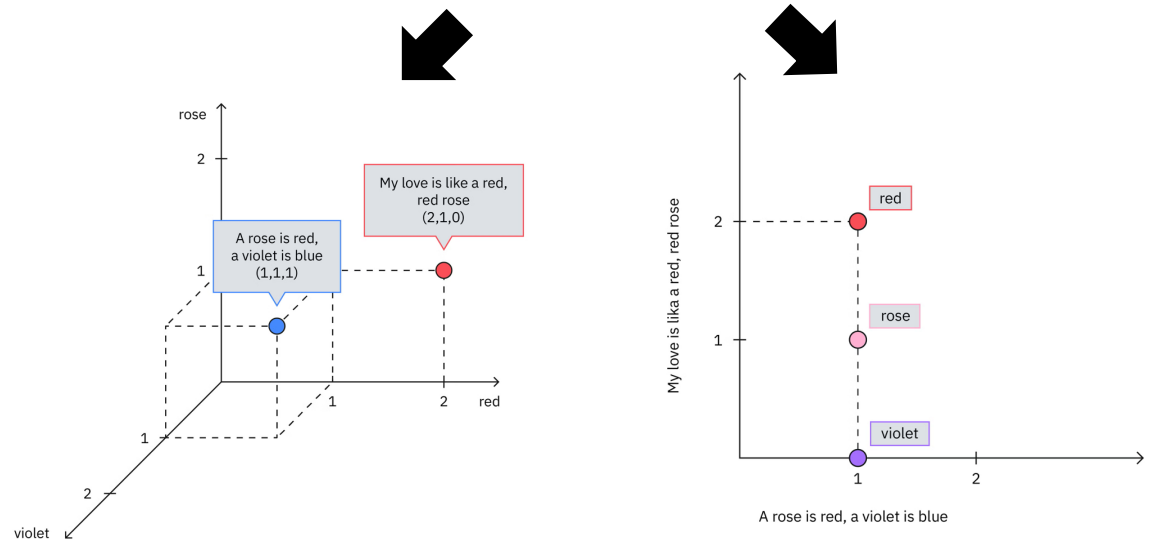
The painting depicts a dog in a humorous situation, where it appears to be wearing a woman's dress and standing in front of a landscape. Although the artist of the painting is not explicitly identified, it could be a reference to the famous artist Da Vinci, known for his artistic works, including the Mona Lisa. The painting is likely a creative and comical take on the traditional portrait style, replacing a human figure with a dog dressed in a woman's clothing, making it an interesting and amusing visual piece.

Bag of Word

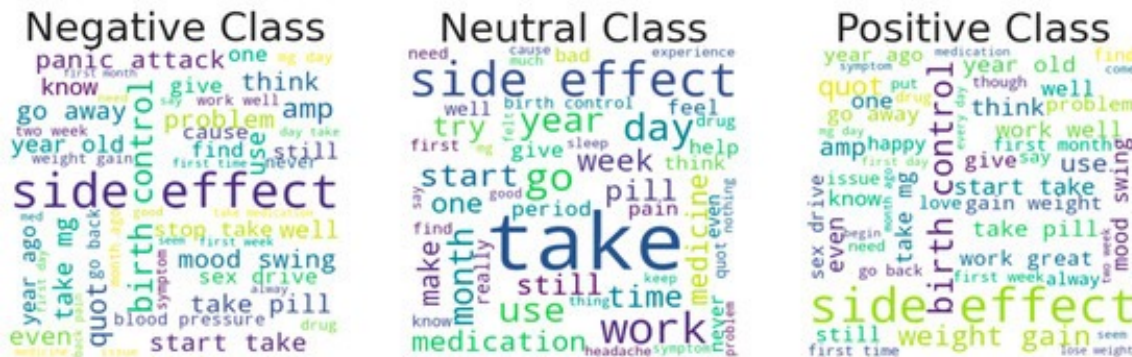
Bag-of-Words



Document 1: *A rose is red, a violet is blue*
Document 2: *My love is like a red, red rose*



EX) Online Drug Reviews



While bag of words counts only the number of times a word appears in one document, TF-IDF accounts for the word's prevalence throughout every document in a text set. TF-IDF is represented in the equation:

$$TF-IDF = (\text{word count in document } x) \times \frac{N}{n}$$

N equals the total number of documents in the text set, and n equals the number of documents in which a given word appears.

Problem: Out-of-Vocabulary (OOV)

Vocabulary: low, lower, newest, widest

If 'lowest' as input

Solution1 to OOV: Byte Pair Encoding (BPE)

Vocabulary: low, lower, newest, widest

If 'lowest' as input

dictionary low : 5, lower : 2, newest : 6, widest : 3

vocabulary l, o, w, e, r, n, s, t, i, d

**Iter 1 – Merge two characters as one that are most frequent
(e, s) 9 -> es**

dictionary update! low : 5, lower : 2, newest : 6, widest : 3

vocabulary update! l, o, w, e, r, n, s, t, i, d, es

Solution1 to OOV: Byte Pair Encoding (BPE)

Vocabulary: low, lower, newest, widest

If 'lowest' as input

dictionary update! l o w : 5, l o w e r : 2, n e w e s t : 6, w i d e s t : 3

vocabulary update! l, o, w, e, r, n, s, t, i, d, e s

Iter 2 – Merge (es, t) whose frequency is 9

dictionary update! l o w : 5, l o w e r : 2, n e w e s t : 6, w i d e s t : 3

vocabulary update! l, o, w, e, r, n, s, t, i, d, e s, e s t

Solution1 to OOV: Byte Pair Encoding (BPE)

Vocabulary: low, lower, newest, widest

If 'lowest' as input

dictionary update! l o w : 5, l o w e r : 2, n e w e s t : 6, w i d e s t : 3

vocabulary update! l, o, w, e, r, n, s, t, i, d, e s, e s t

Iter3 – Merge (l, o) that are most frequent characters (frequency 7)

dictionary update! l o w : 5, l o w e r : 2, n e w e s t : 6, w i d e s t : 3

vocabulary update! l, o, w, e, r, n, s, t, i, d, e s, e s t, l o

Solution1 to OOV: Byte Pair Encoding (BPE)

Vocabulary: low, lower, newest, widest

If 'lowest' as input

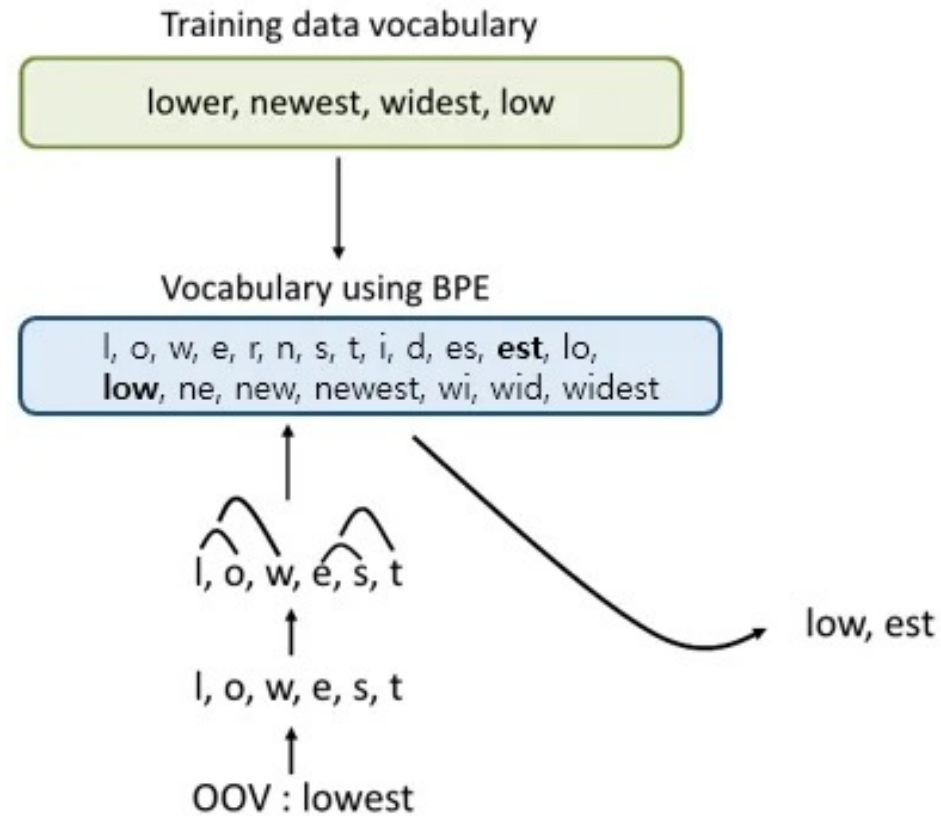
After 10 time iterations, we can obtain as below:

dictionary update! low : 5, low e r : 2, newest : 6, widest : 3

vocabulary update! l, o, w, e, r, n, s, t, i, d, es, **est**, lo, **low**, ne, new, newest, wi, wid, widest

- ✓ Machine will split 'lowest' as 'l, o, w, e, s, t'.
- ✓ Then, the machine will find 'low' and 'est' based on the updated vocabulary
- ✓ That's, 'lowest' will be encoded with both 'low' and 'est'
- ✓ Therefore, since the two words, 'low' and 'est', exist in the vocabulary, 'lowest' is not OOV

Solution1 to OOV: Byte Pair Encoding (BPE)



Solution2 to OOV: WordPiece Tokenization

Vocabulary: low, lower, newest, widest

If 'lowest' as input

```
# dictionary  
l o w : 5,  
l o w e r : 2,  
n e w e s t : 6,  
w i d e s t : 3
```

```
# vocabulary  
l, ##o, ##w  
l, ##o, ##w, ##e, ##r  
n, ##e, ##w, ##e, ##s, ##t  
w, ##i, ##d, ##e, ##s, ##t
```



means characters that do not start a word

Solution2 to OOV: WordPiece Tokenization

Vocabulary: low, lower, newest, widest

If 'lowest' as input

dictionary

low : 5, → 5

lower : 2, → 2

newest : 6, → 6

widest : 3 → 3

split

l, ##o, ##w

l, ##o, ##w, ##e, ##r

n, ##e, ##w, ##e, ##s, ##t

w, ##i, ##d, ##e, ##s, ##t

vocabulary

l, ##o, ##w, ##e, ##r

n, ##s, ##t, w, ##i, ##d

$$\text{Score}(l, o) = \frac{\text{freq}("lo")}{\text{freq}("l") \times \text{freq}("o")} = \frac{7}{7 \times 7} = 0.143$$

Solution2 to OOV: WordPiece Tokenization

Vocabulary: low, lower, newest, widest

If 'lowest' as input

dictionary

low : 5, —————→ 5

lower : 2, —————→ 2

newest : 6, —————→ 6

widest : 3 —————→ 3

split

l, ##o, ##w

l, ##o, ##w, ##e, ##r

n, ##e, ##w, ##e, ##s, ##t

w, ##i, ##d, ##e, ##s, ##t

vocabulary

l, ##o, ##w, ##e, ##r

n, ##s, ##t, w, ##i, ##d

$$\text{Score}(e, s) = \frac{\text{freq}("es")}{\text{freq}("e") \times \text{freq}("s")} = \frac{9}{17 \times 9} = 0.058$$

Solution2 to OOV: WordPiece Tokenization

Vocabulary: low, lower, newest, widest

If 'lowest' as input

dictionary

low : 5, —————→ 5

lower : 2, —————→ 2

newest : 6, —————→ 6

widest : 3 —————→ 3

split

l, ##o, ##w

l, ##o, ##w, ##e, ##r

n, ##e, ##w, ##e, ##s, ##t

w, ##i, ##d, ##e, ##s, ##t

vocabulary

l, ##o, ##w, ##e, ##r

n, ##s, ##t, w, ##i, ##d

$$\text{Score}(w, e) = \frac{\text{freq}(\text{"we"})}{\text{freq}(\text{"w"}) \times \text{freq}(\text{"e"})} = \frac{8}{11 \times 11} = 0.066$$

Solution2 to OOV: WordPiece Tokenization

Vocabulary: low, lower, newest, widest

If 'lowest' as input

dictionary

low : 5, → 5

lower : 2, → 2

newest : 6, → 6

widest : 3 → 3

split

l, ##o, ##w

l, ##o, ##w, ##e, ##r

n, ##e, ##w, ##e, ##s, ##t

w, ##i, ##d, ##e, ##s, ##t

vocabulary

l, ##o, ##w, ##e, ##r

n, ##s, ##t, w, ##i, ##d

1. Calculate the scores for all pairs

$$\text{Score}(w, e) = \frac{\text{freq}(\text{pair})}{\text{freq}(\text{first element}) \times \text{freq}(\text{second element})}$$

Solution2 to OOV: WordPiece Tokenization

Vocabulary: low, lower, newest, widest

If 'lowest' as input

dictionary

low : 5, —————→ 5

lower : 2, —————→ 2

newest : 6, —————→ 6

widest : 3 —————→ 3

split

l, ##o, ##w

l, ##o, ##w, ##e, ##r

n, ##e, ##w, ##e, ##s, ##t

w, ##i, ##d, ##e, ##s, ##t

vocabulary

l, ##o, ##w, ##e, ##r

n, ##s, ##t, w, ##i, ##d, lo

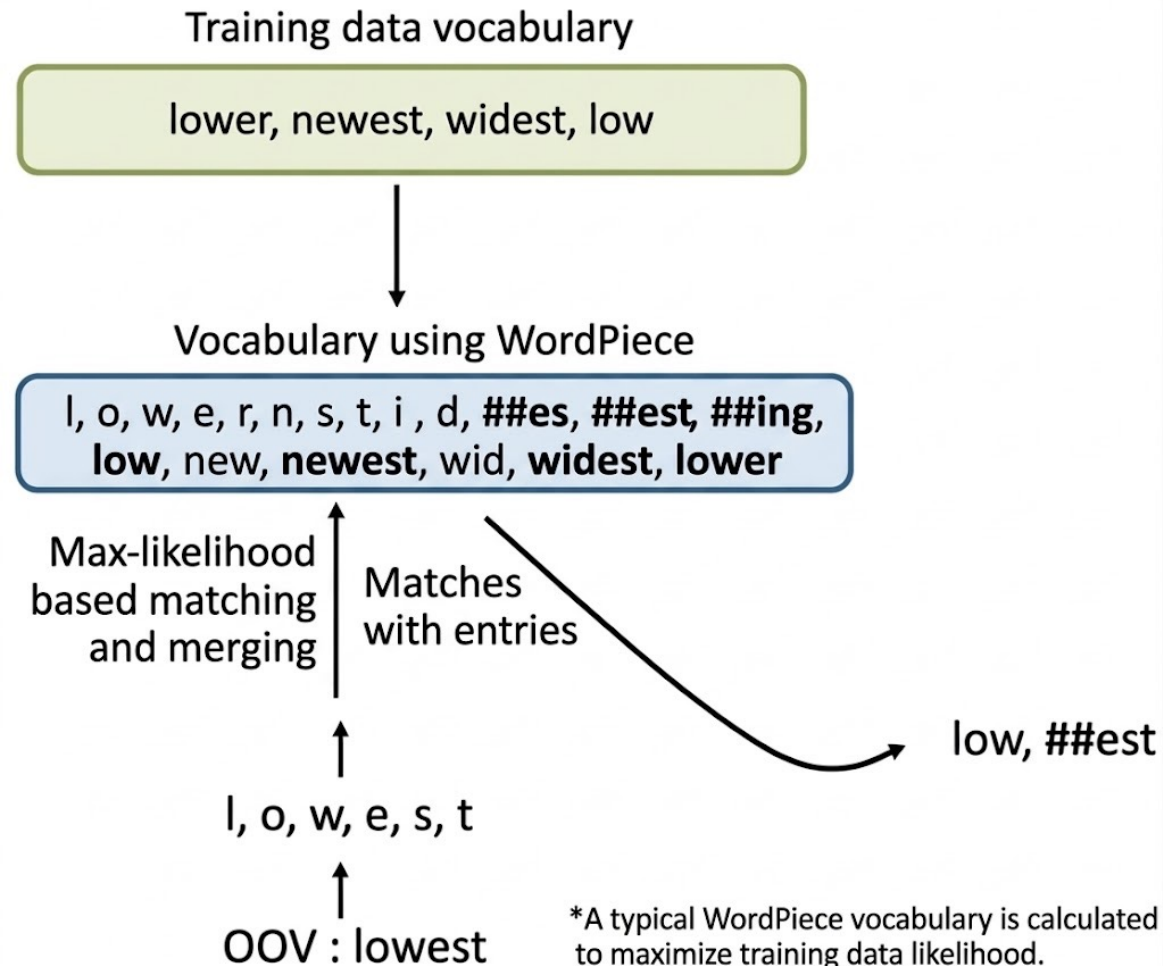
2. Update word with the maximum score into vocabulary

$$\text{Score}(l, o) = \frac{7}{7 \times 7} = 0.143$$

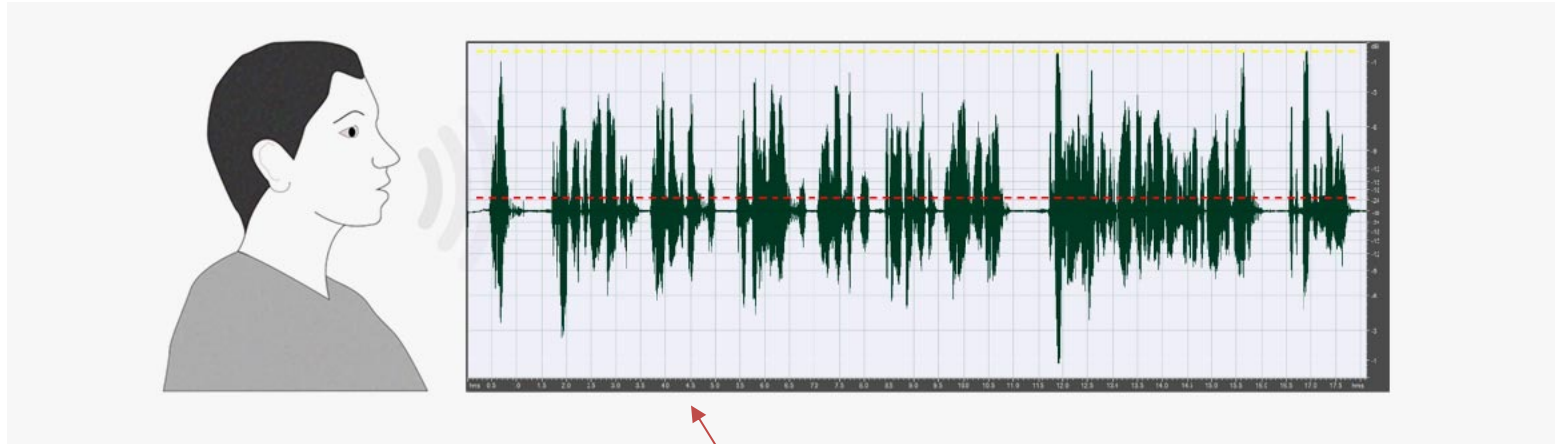
$$\text{Score}(e, s) = \frac{9}{17 \times 9} = 0.058$$

$$\text{Score}(w, e) = \frac{8}{11 \times 11} = 0.066$$

Solution2 to OOV: WordPiece Tokenization



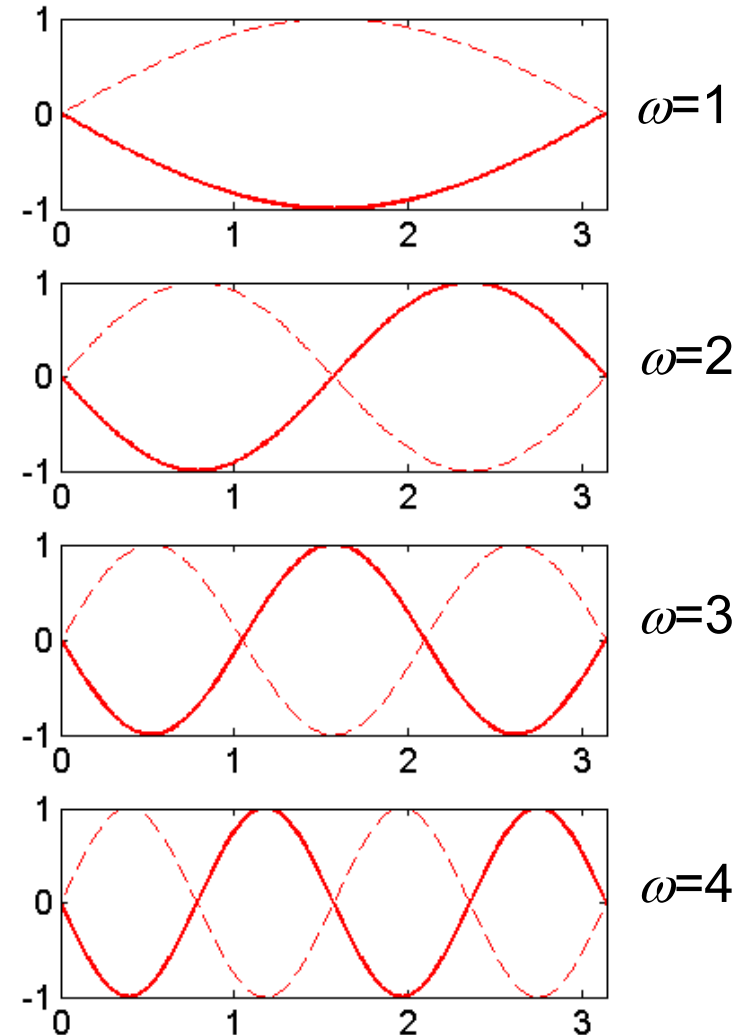
Question: How to handle time-series data?



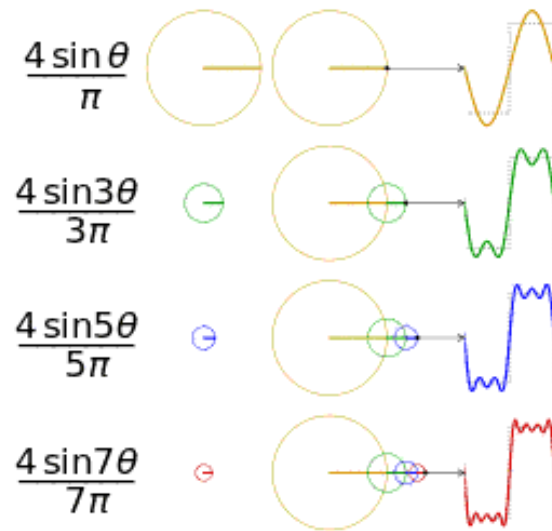
Human voice is a very complex time-series signal

Question: How to handle time-series data?

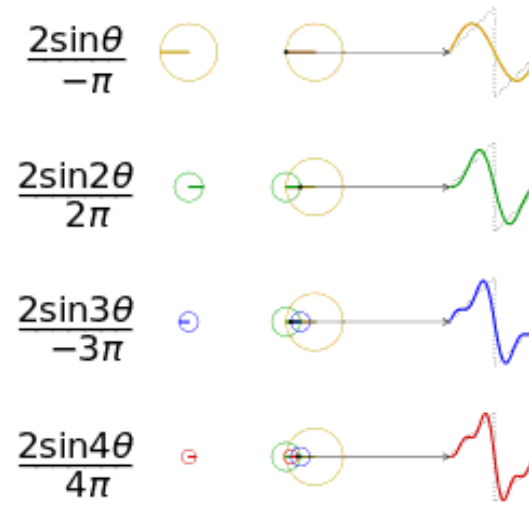
- The idea of using trigonometric sums was used to predict astronomical events
- Euler studied vibrating strings, ~1750, which are signals where linear displacement was preserved with time.
- Fourier described how such a series could be applied and showed that a periodic signals can be represented as the integrals of sinusoids that are not all harmonically related
- Now widely used to understand the structure and frequency content of arbitrary signals



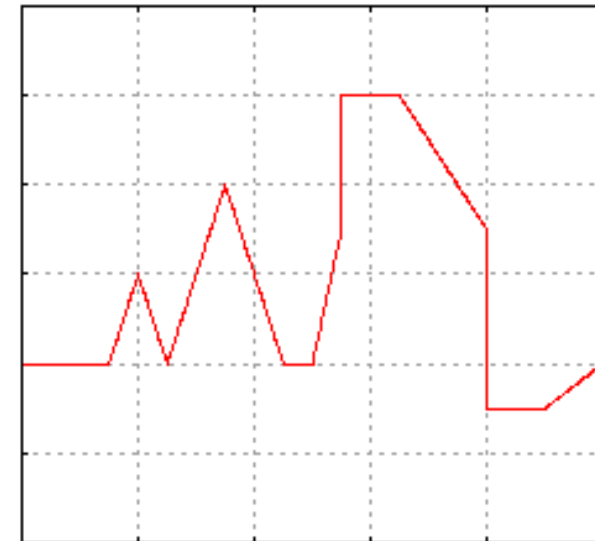
Question: How to handle time-series data?



A Rectangular Pulse
(Square Wave)



A Sawtooth Wave

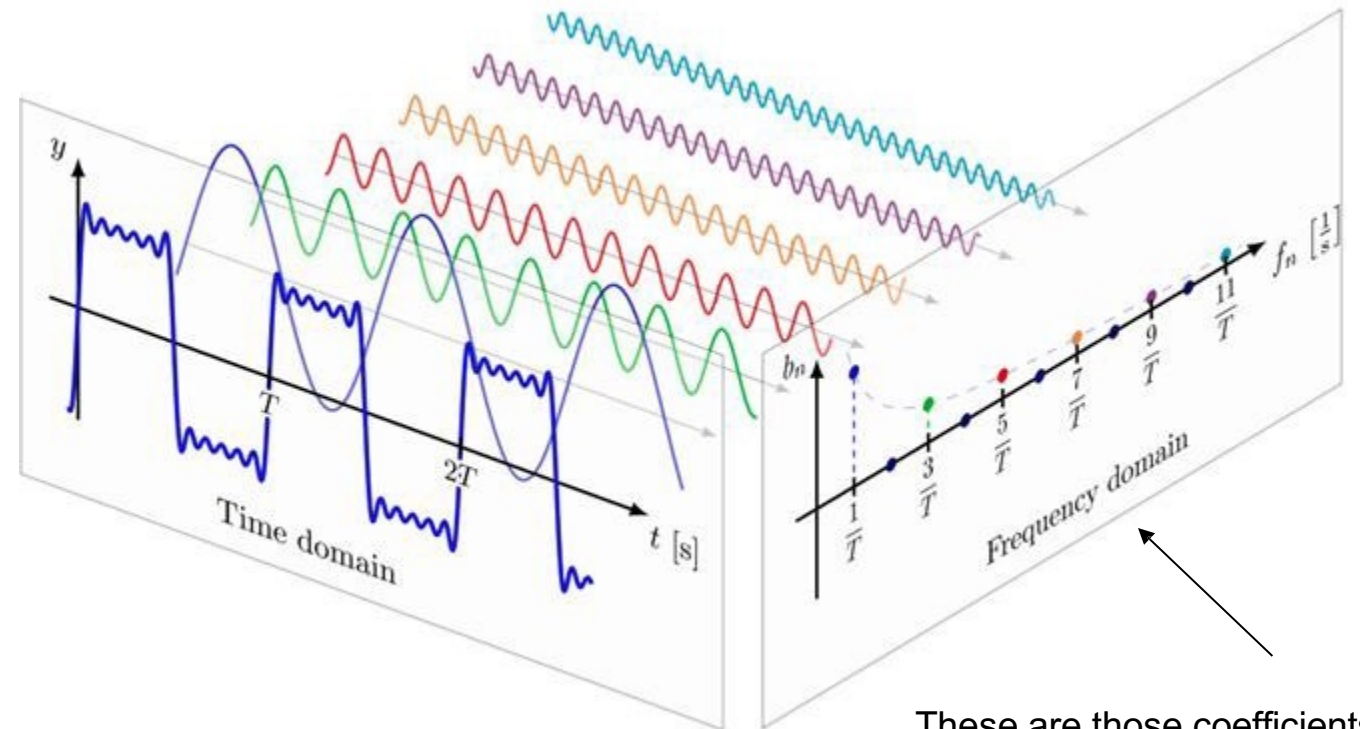


Any Signal

Time domain to Frequency domain

Any univariate function can be rewritten as a weighted sum of sines and cosines of different frequencies

i.e., if we weighted sum across the different frequencies, we reconstruct the original signal



Fourier Transform

We will be referring to functions of time and their Fourier transforms.

A signal $x(t)$ and its Fourier transform $X(j\omega)$ are related by the **Fourier transform synthesis** and **analysis** equations

$$X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t} dt = F\{x(t)\}$$

and

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(j\omega)e^{j\omega t} d\omega = F^{-1}\{X(j\omega)\}$$

We will refer to $x(t)$ and $X(j\omega)$ as a **Fourier transform pair** with the notation

$$x(t) \overset{F}{\longleftrightarrow} X(j\omega)$$

Discrete Fourier Transform

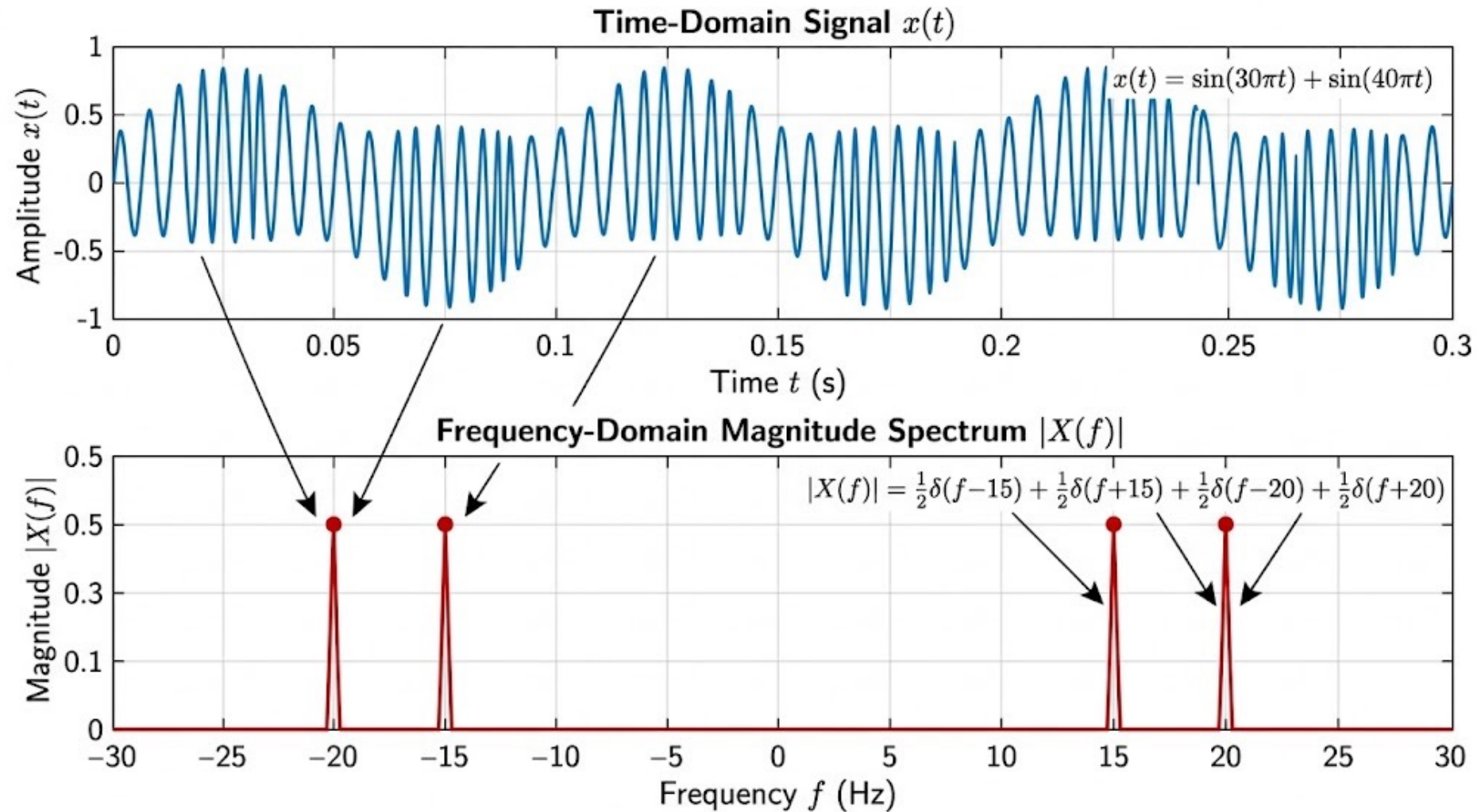
The DT Fourier transform **analysis** and **synthesis** equations are therefore:

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$
$$x[n] = \frac{1}{2\pi} \int_{2\pi} X(e^{j\omega})e^{j\omega n} d\omega$$

The function $X(e^{j\omega})$ is referred to as the **discrete-time Fourier transform** and the pair of equations are referred to as the **Fourier transform pair**

$X(e^{j\omega})$ is sometimes referred to as the **spectrum** of $x[n]$ because it provides us with information on how $x[n]$ is composed of **complex exponentials at different frequencies**

Example of Fourier Transform



Gain and Phase Transfer Function Analysis

For a complex number/function, we can represent it in polar form by calculating the **magnitude (gain)** and **angle (phase)**:

$$H(j\omega) = |H(j\omega)| e^{j\angle H(j\omega)}$$

In filter/system analysis and design, we're interested in how certain frequencies are magnified or suppressed –

These are of particular interest by plotting the system properties (amplitude/phase) against frequency:

- Frequency shaping
- Frequency selection
- Low pass
- High pass

The Concept of Filtering

With a bit of work (next slide) it can show that:

$$y(t) = h(t) * x(t) \xleftrightarrow{F} Y(j\omega) = H(j\omega)X(j\omega)$$

Therefore, to apply **convolution in the frequency domain**, we just have to multiply the two functions.

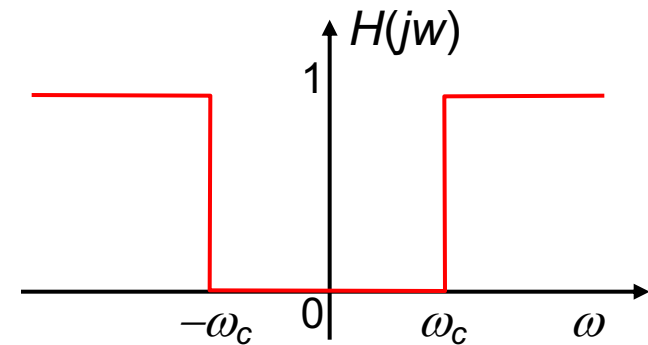
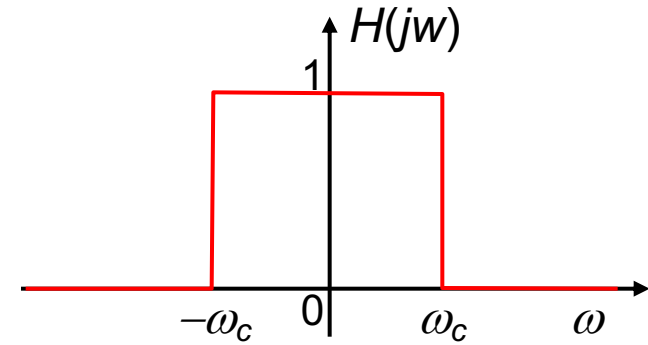
To solve for the differential/convolution equation using Fourier transforms:

1. Calculate Fourier transforms of $x(t)$ and $h(t)$
2. Multiply $H(j\omega)$ by $X(j\omega)$ to obtain $Y(j\omega)$
3. Calculate the inverse Fourier transform of $Y(j\omega)$

Multiplication in the frequency domain corresponds to convolution in the time domain and vice versa.

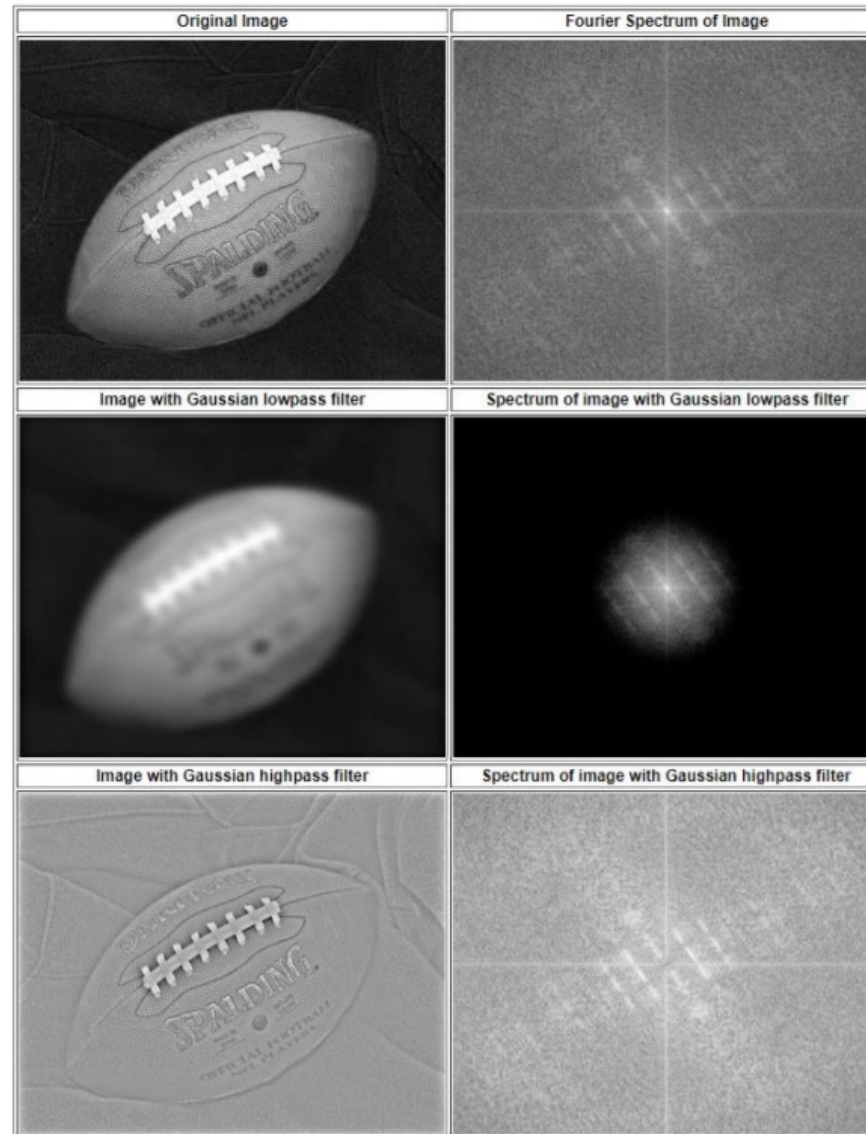
The Concept of Filtering

- **Frequency selection filters** are specially designed to accept some frequencies and reject others. Noise in an audio recording can be removed by low pass filtering, multiple communication signals can be encoded at different frequencies and then recovered by selecting particular frequencies
- **Low pass filters** are designed to reject/attenuate high frequency “noise” while passing on the low frequencies
- **High pass filters** are designed to reject/attenuate low frequency signal components while passing on high frequency

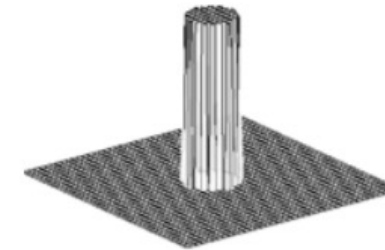


Filtering – 2D Image Signal

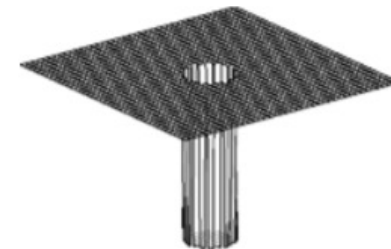
Image - Fourier transform



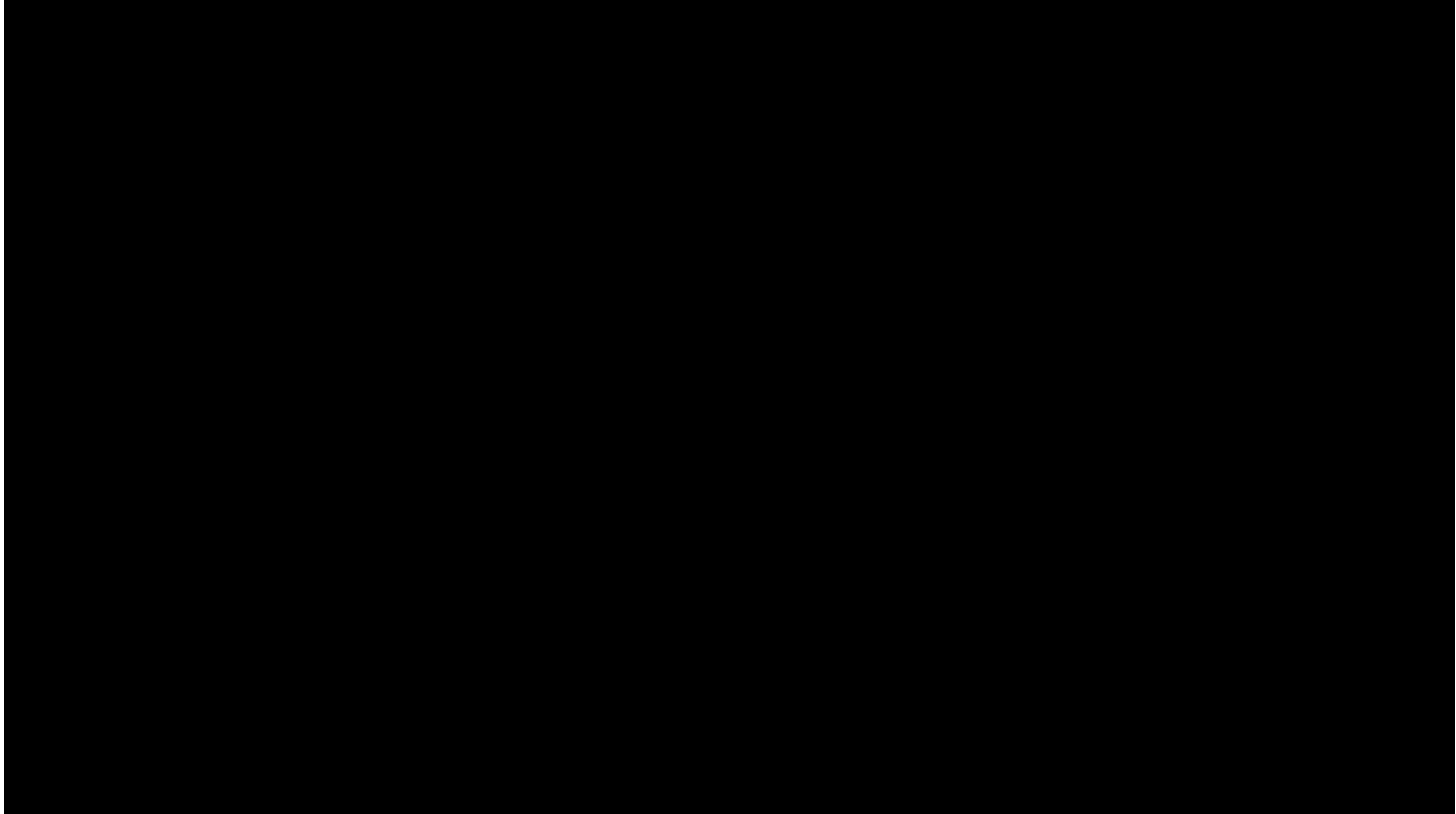
Low-pass filter



High-pass filter

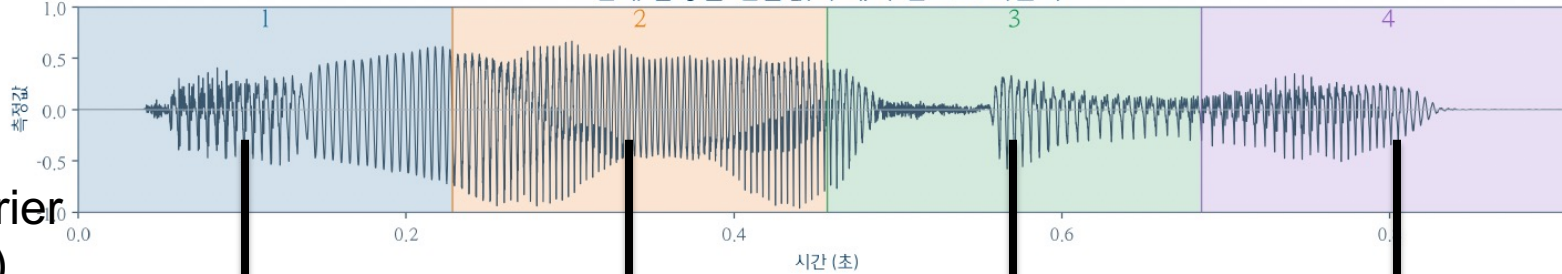


Filtering – 1D Voice Signal



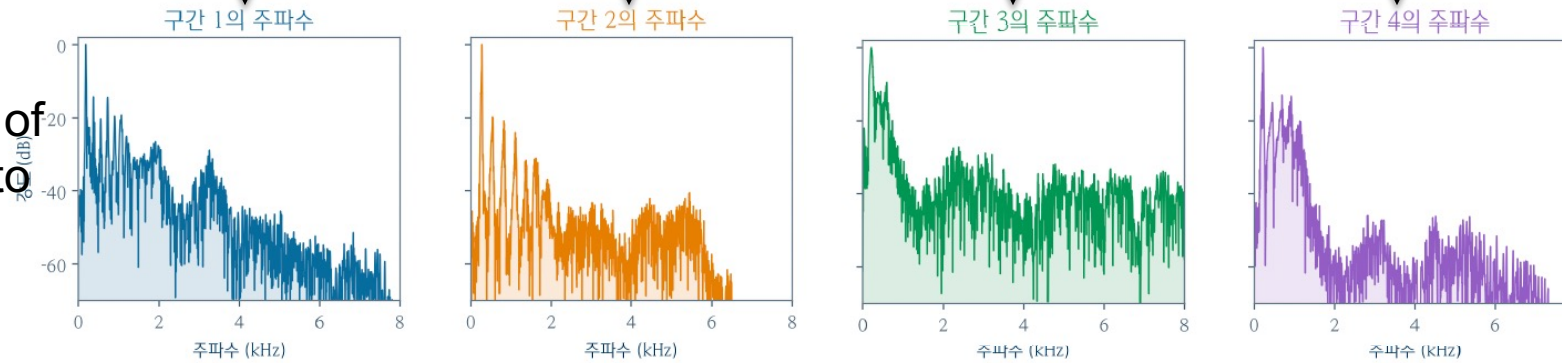
Spectrogram

전체 음성을 빈틈없이 네 구간으로 나눈다



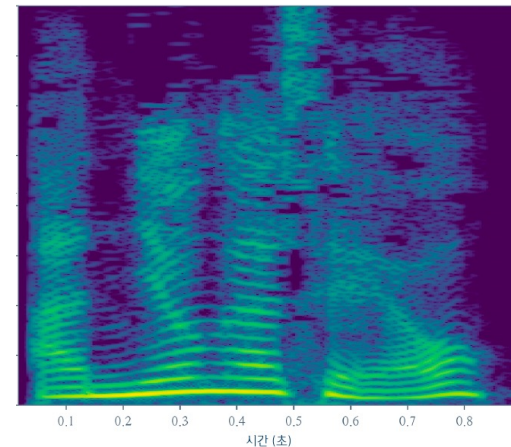
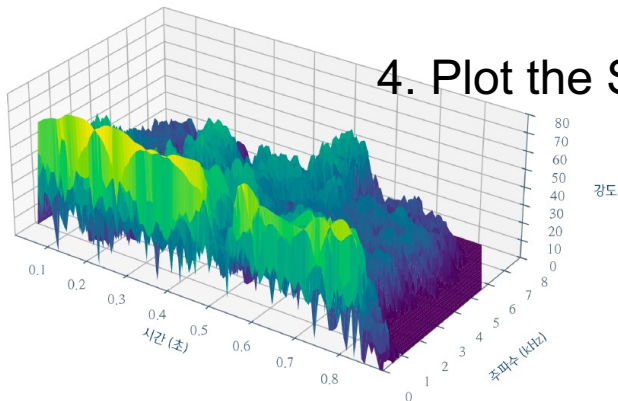
1. Short-time Fourier Transform (STFT)

2. Absolute value of the STFT results to compute their magnitude

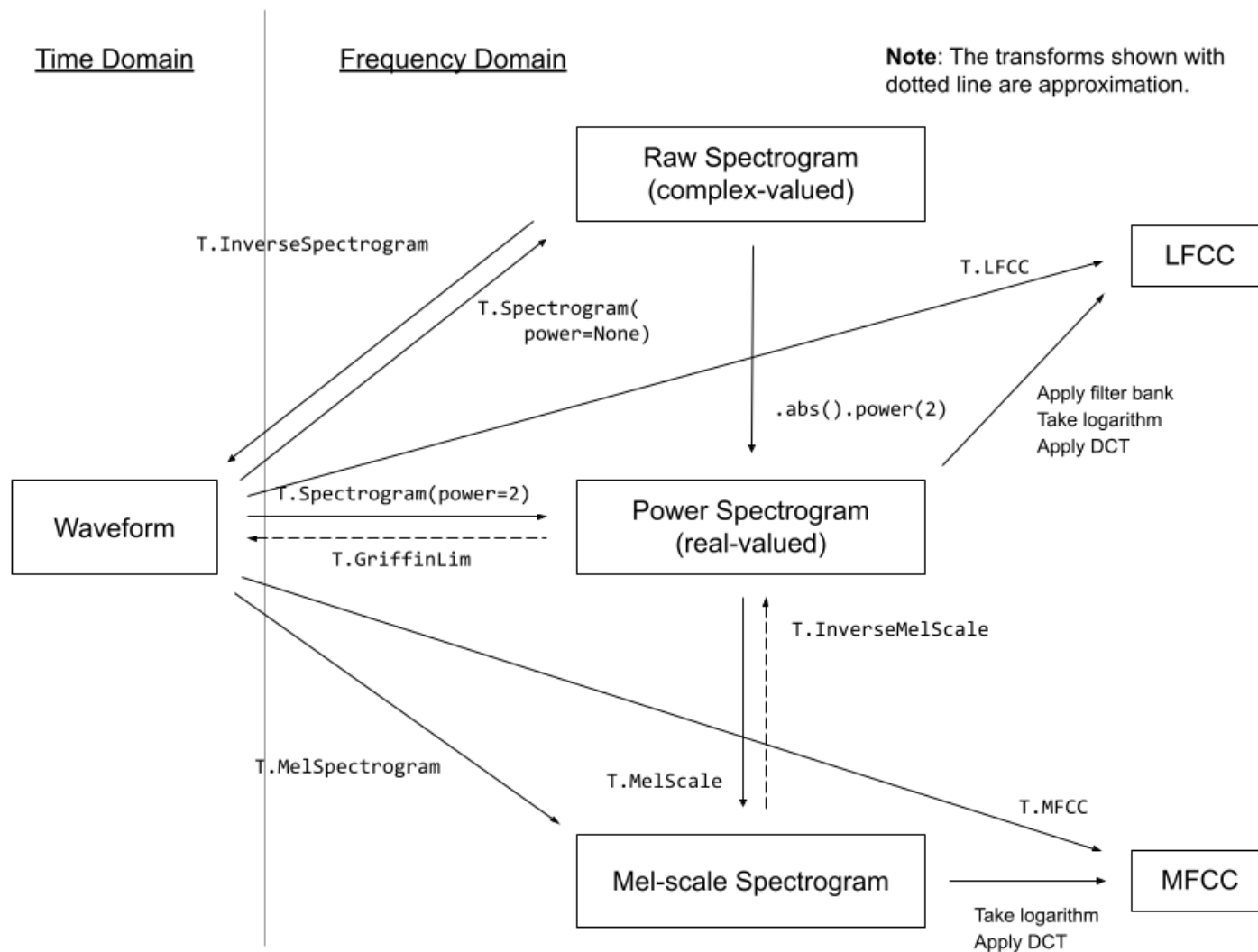


3. Convert to Decibels (dB): Scale the magnitude to decibels to better match human hearing.

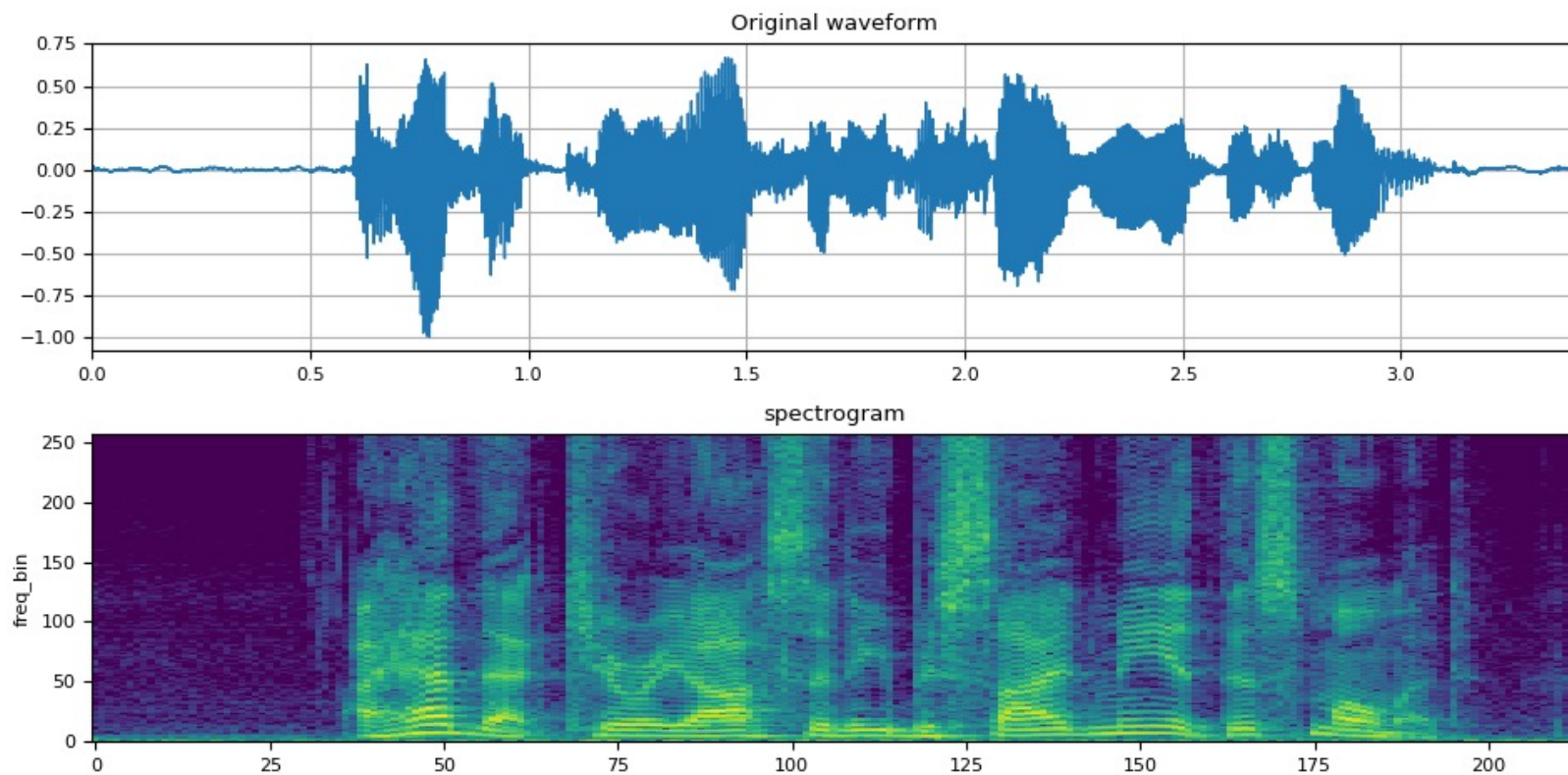
4. Plot the Spectrogram



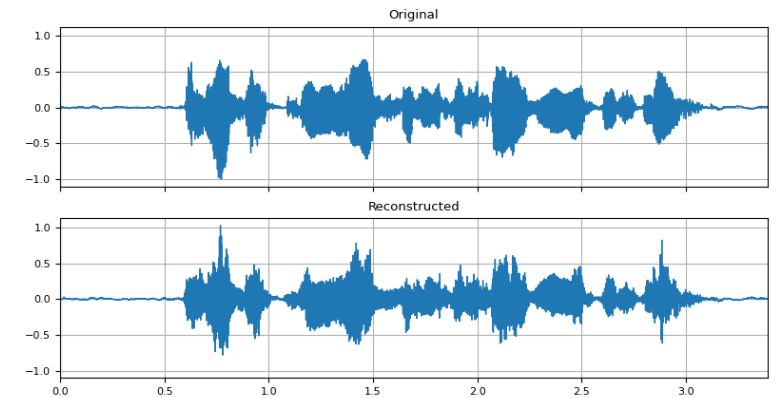
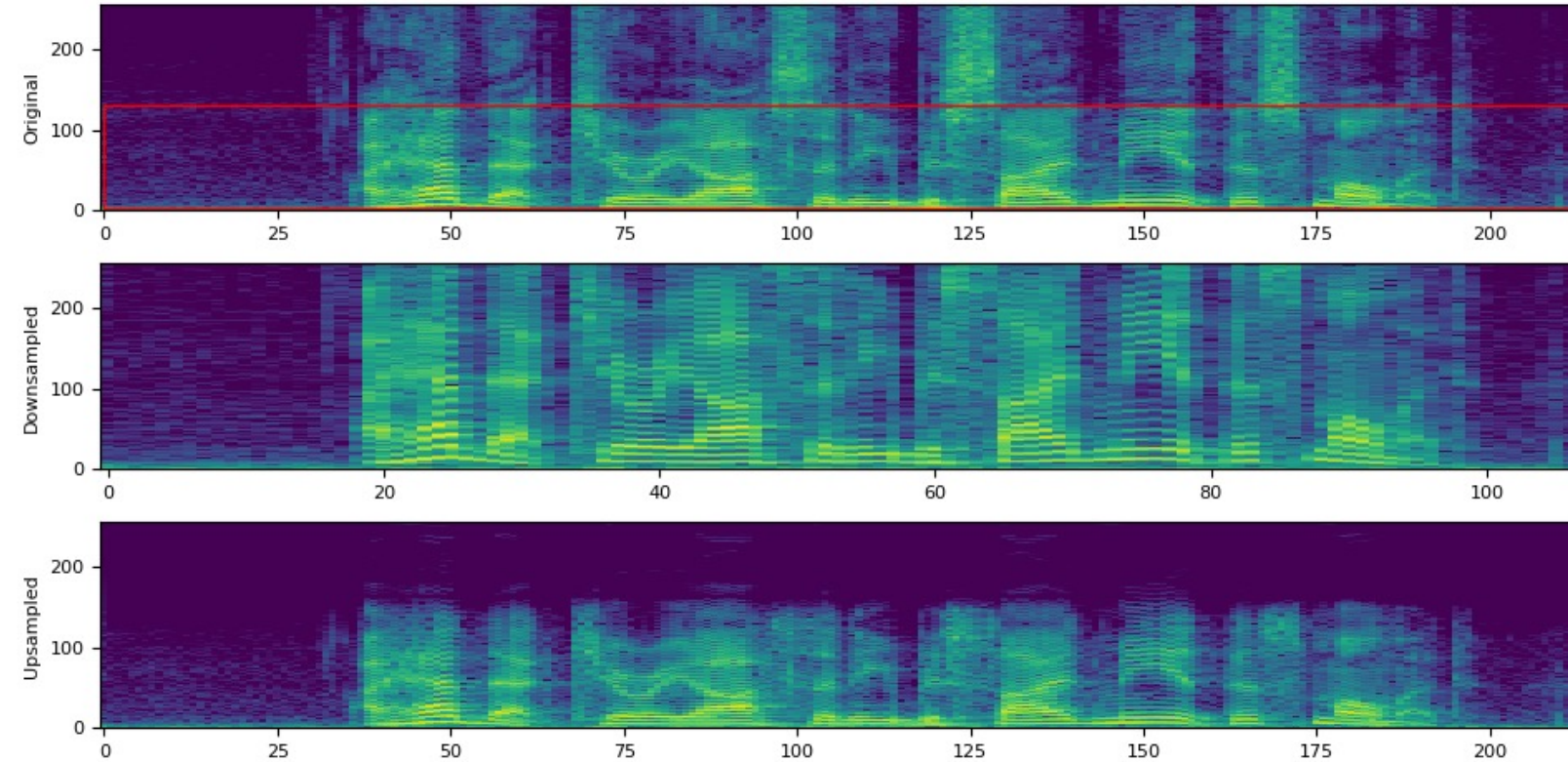
Spectrogram



Voice Processing using Spectrogram



Voice Processing using Spectrogram



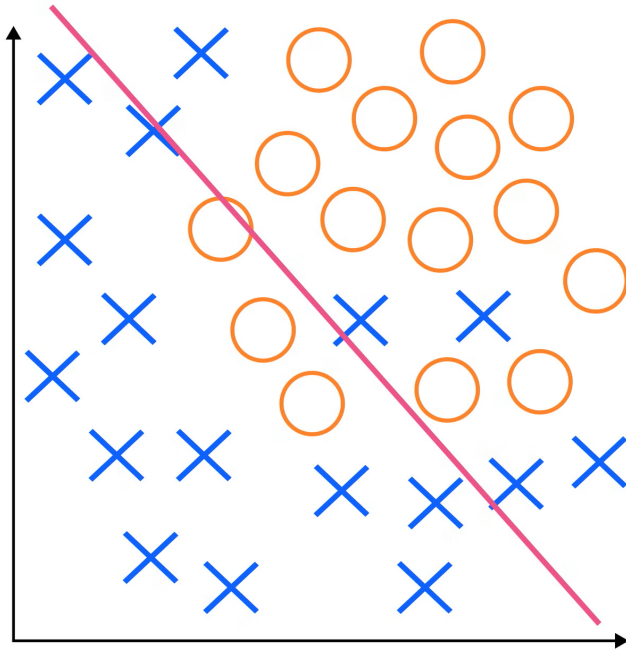
- We can process the audio signal like the image file
- More techniques will be introduced later, especially data augmentation

Question: What are factors that make models' performance worse w.r.t. dataset?



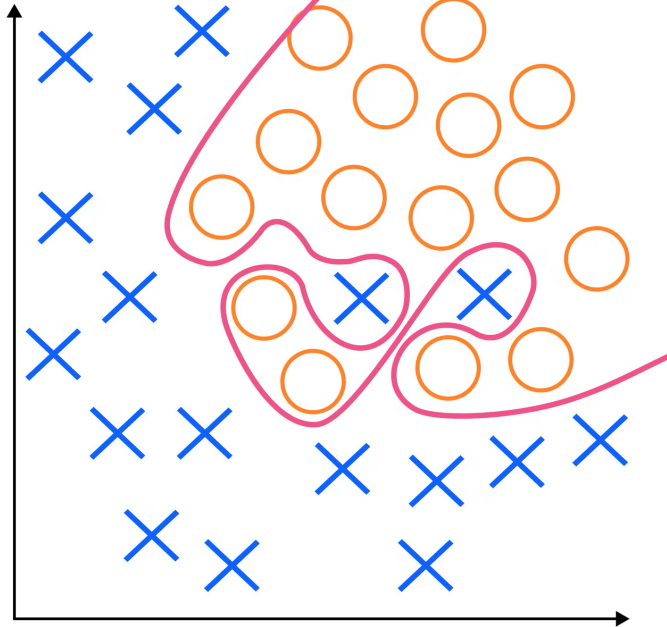
Underfitting and Overfitting

Underfitting



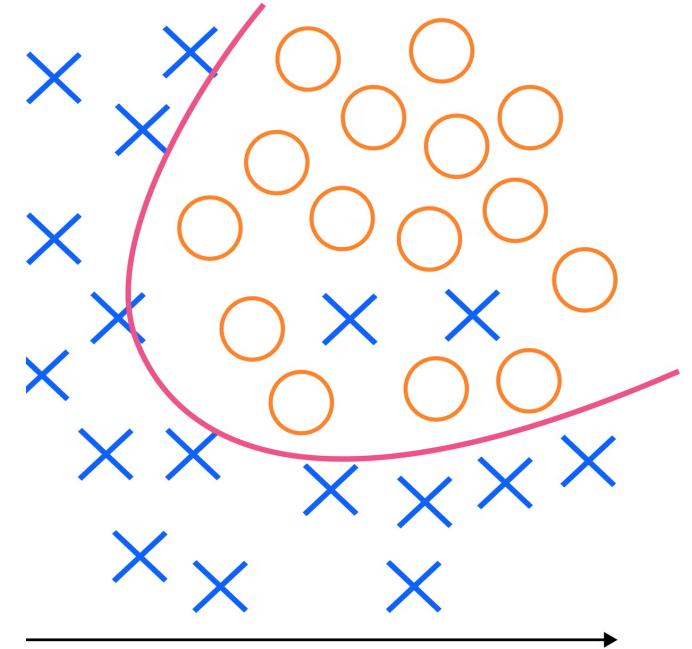
- **Errors are consistently high** across training and testing data sets.
- **performs poorly** on training data and testing data because it fails to capture the dominant patterns in the data set.
- Engineers typically identify underfitting through consistently **poor performance** across both data sets.

Overfitting

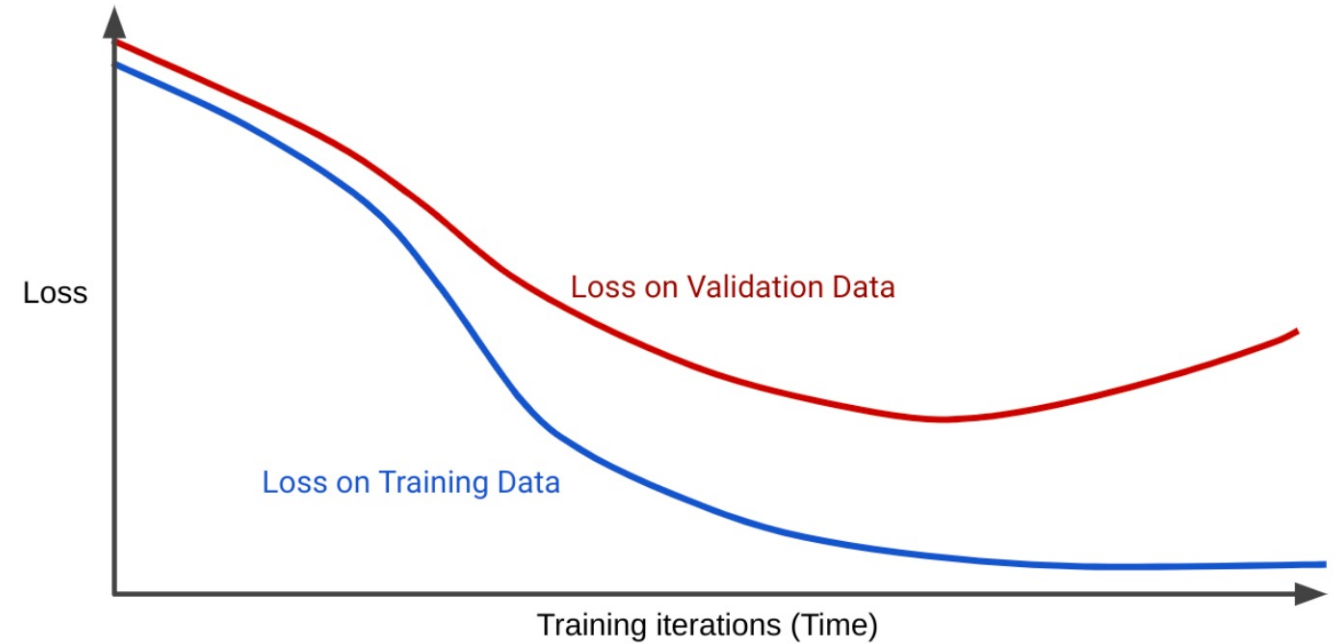
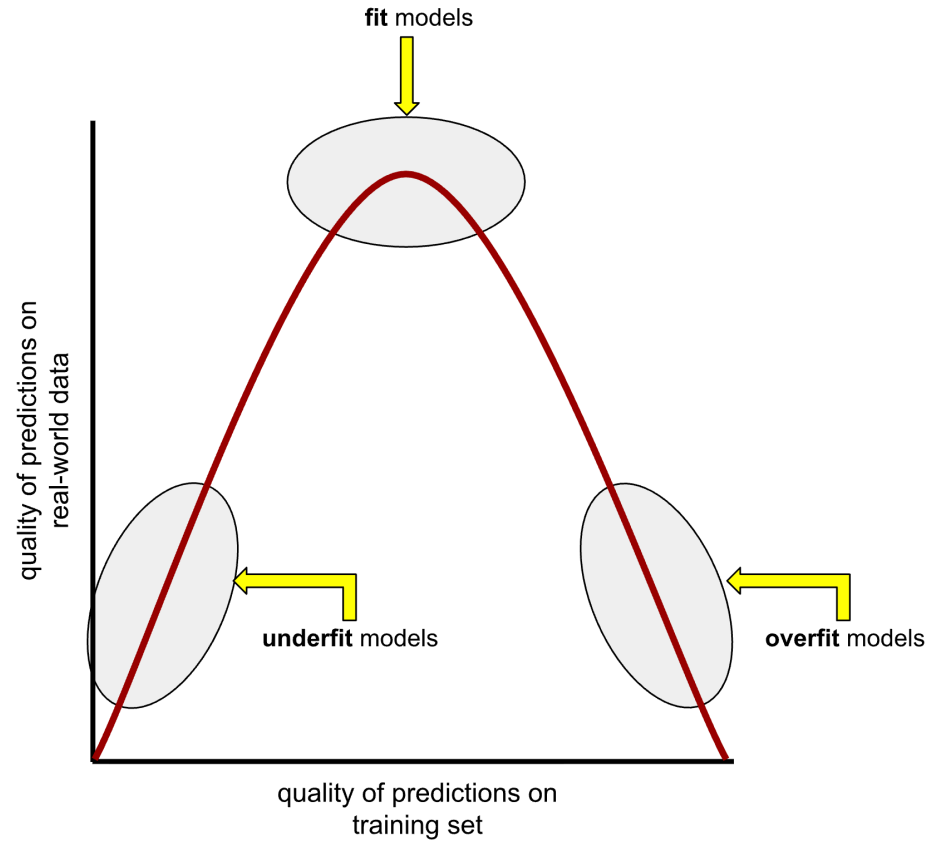


- **Training error is low, but testing error is significantly higher.**
- can result in high model accuracy on training data but low accuracy on new data due to memorization instead of generalization.
- happens when engineers use a machine learning **model with too many parameters or layers**

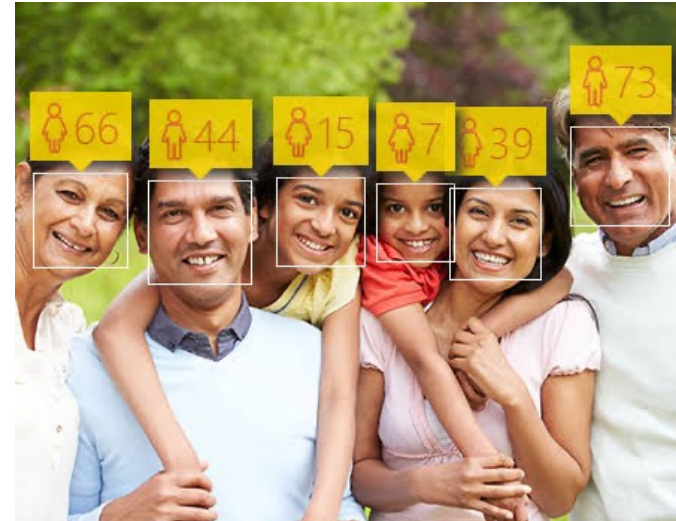
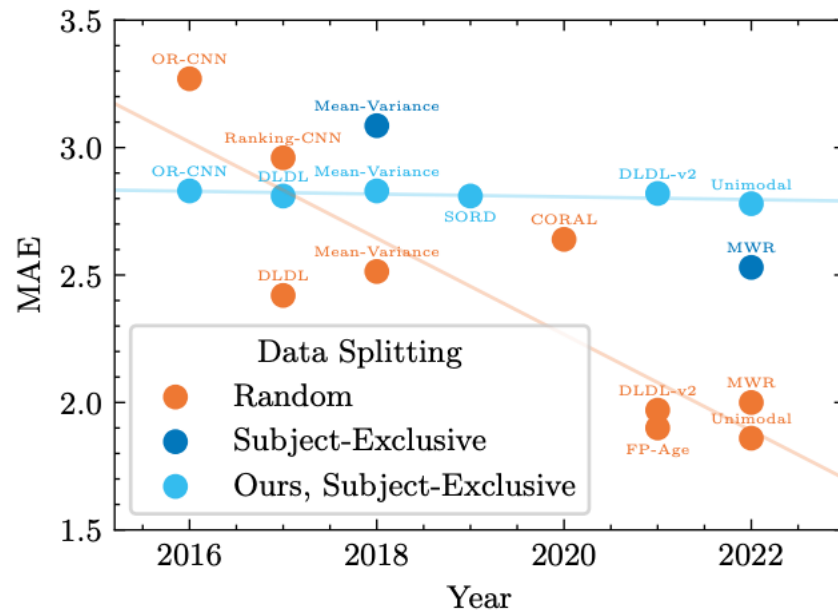
Appropriate fitting



Underfitting and Overfitting



Data Leakage



The latest CVPR'24 research paper found that the reported continuous **performance improvements** over the past decade using specialized methods are in fact **negligible** when accounting for other factors, such as **data leakage**.

Data Leakage

Data leakage occurs when information from outside the training data is used to build the model.

Formally: Training procedure has access to information from the test set

Consequences:

- ✓ Overly optimistic performance estimates
- ✓ Poor generalization to new data
- ✓ Irreproducible scientific results

Common Source:

Scaling or normalization using the full dataset

Feature selection performed before cross-validation

Batch correction using test samples

Reusing test data for model tuning

Key insight: If the test data influences any step of model construction, performance estimates are invalid.

Data Leakage

Incorrect workflow:

- 1 Use all samples to select top k genes
- 2 Cross-validate classifier on selected genes

Why this is wrong:

Feature selection uses label information
Test folds influence which features are chosen

Result: Inflated accuracy, ROC-AUC, PR-AUC

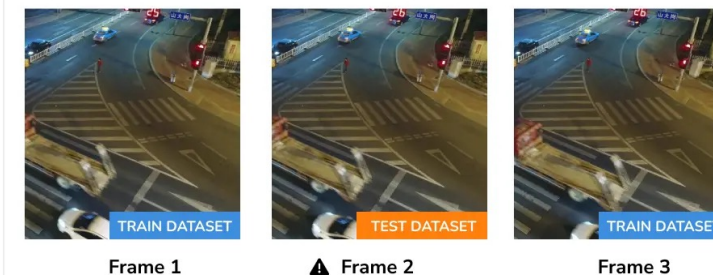
Correct principle:

- ★ Feature selection must be performed inside **the training data only**.

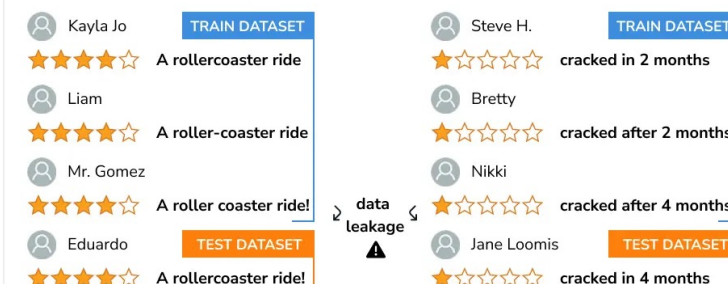
Implementation:

Encapsulate feature selection in a pipeline
Apply selection separately in each CV fold
This ensures that test data remains truly unseen

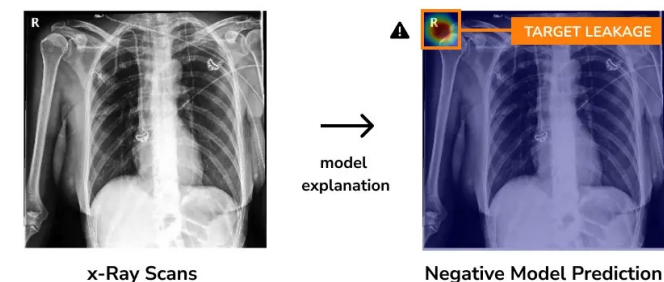
Example: Data Acquisition



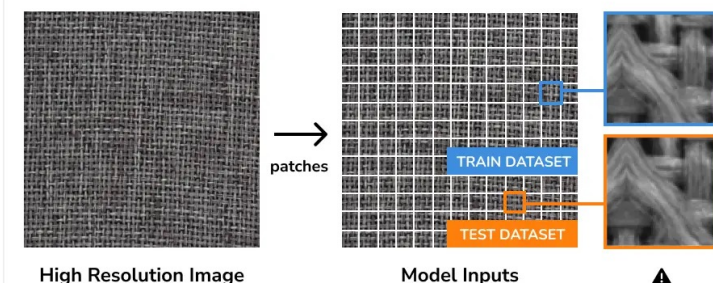
Example: Data Similarity



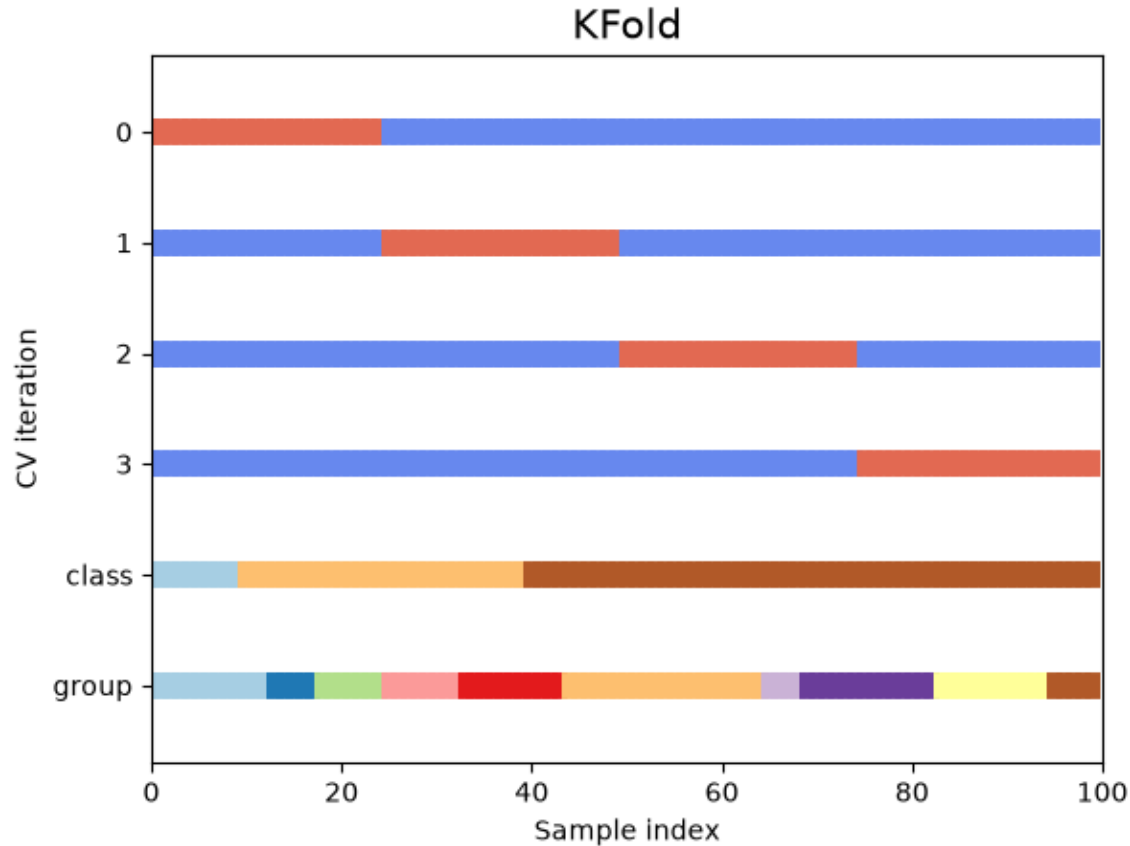
Example: Target Leakage



Example: Data Similarity



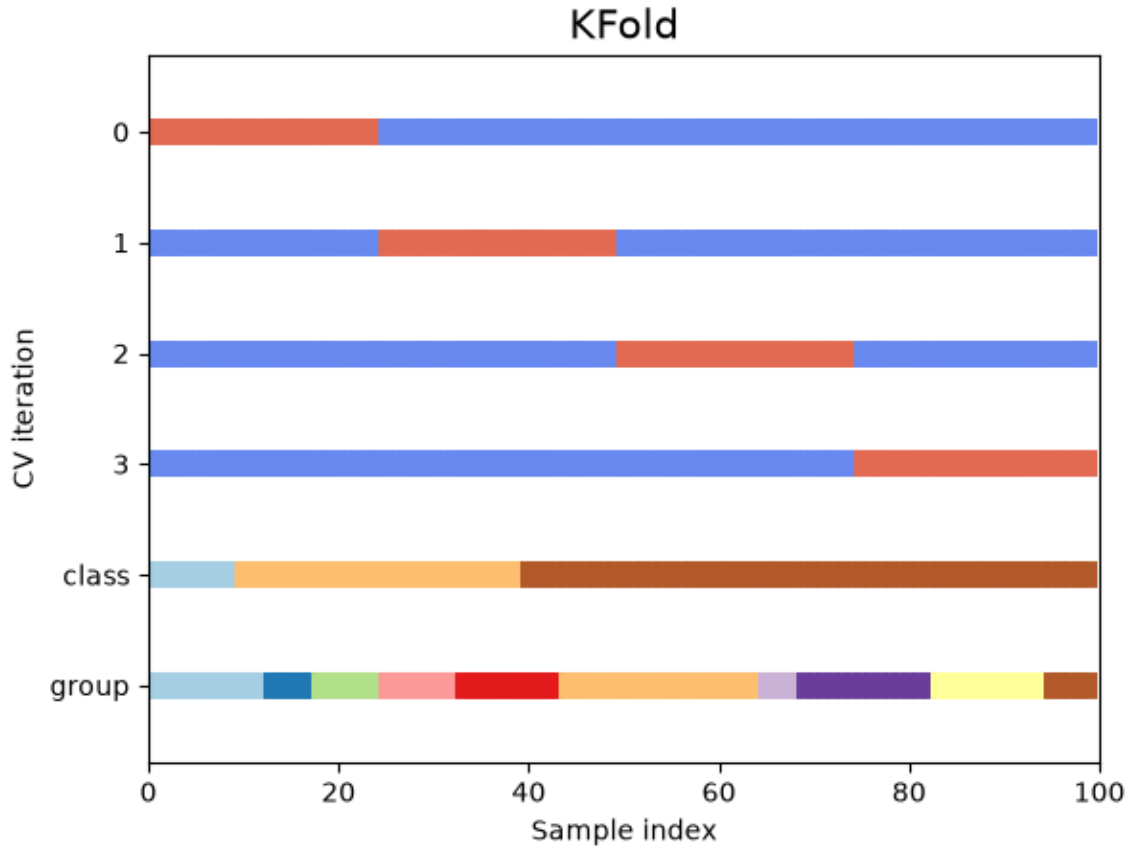
Solution to Data Leakage



CV Iteration (Cross-Validation Split / Fold Round)

- **Location in Plot:** Rows labeled 0, 1, 2, 3 along the Y-axis.
- **Concept:** Represents each discrete split (fold) in the cross-validation cycle. In a 4-fold cross-validation setup, the process executes across four distinct iterations (0 through 3).
- **Visual Meaning:**
 - **Blue bars** mark data indices assigned to the **training set** for that specific fold.
 - **Red bars** mark data indices held out for the **validation set**.
- As the iterations progress, the validation block shifts systematically to ensure every sample serves as a test instance without test-set overlap.

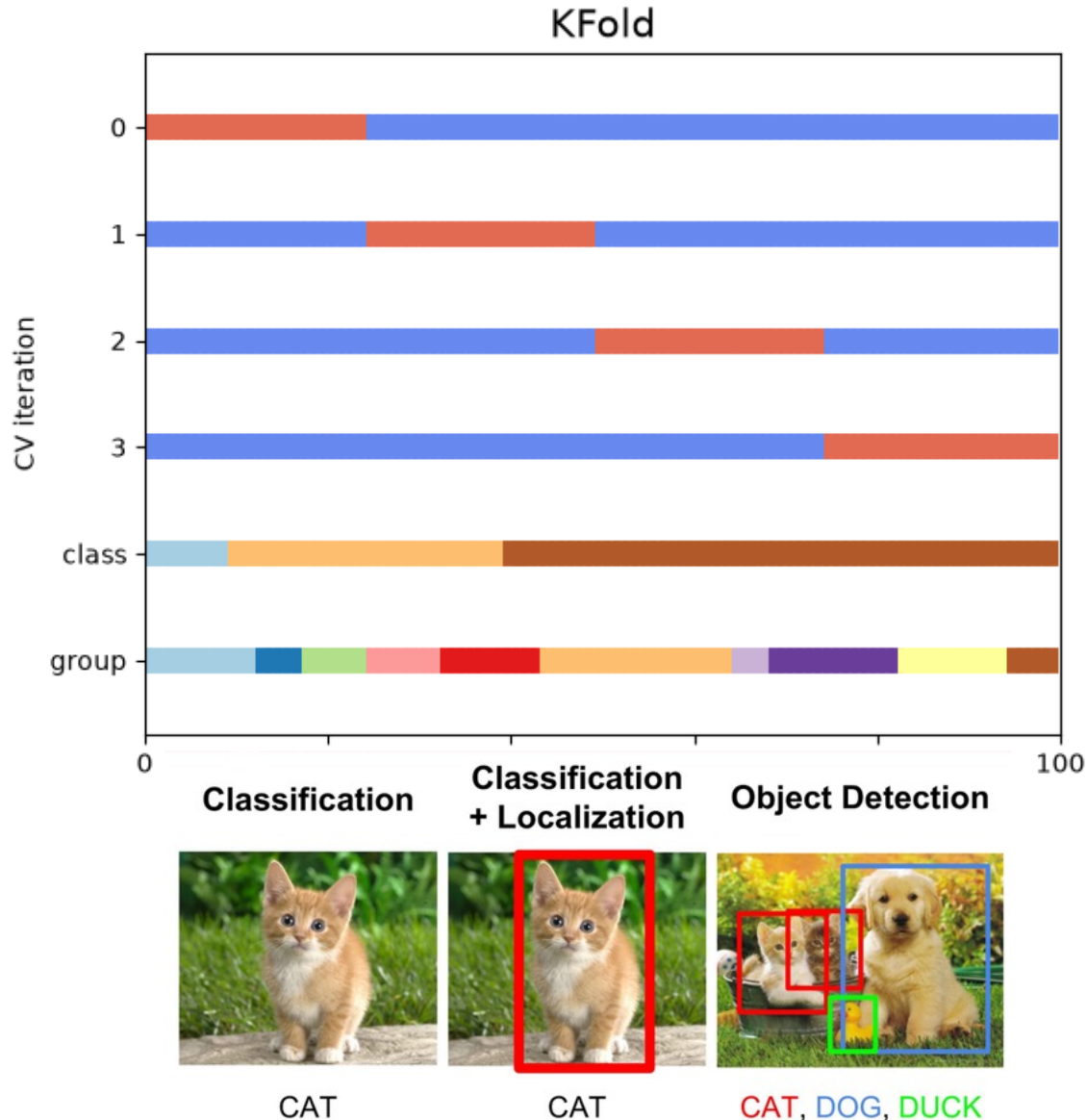
Solution to Data Leakage



Class (Target Labels)

- **Location in Plot:** The row labeled class, positioned directly below the CV iterations.
- **Concept:** The ground-truth category or target variable (y) in a classification task. In the example script, 100 samples are assigned across three classes (0, 1, and 2) with imbalanced distributions of 10%, 30%, and 60%.
- **Why It Matters (Stratification):**
 - Unconstrained splitters like standard KFold or ShuffleSplit sample uniformly at random, which can lead to severe label imbalance across folds or entirely omit minority classes from a test split.
 - Stratified splitters (such as StratifiedKFold) enforce proportional representation, **preserving the overall label ratio** (1:3:6) across both training and test sets in every single fold.

Solution to Data Leakage



Group (Inter-Sample Dependency Identifier)

- **Location in Plot:** The bottom-most row labeled group.
- **Concept:** Metadata indicating that certain samples share a common origin or underlying dependency.
 - Examples: Multiple medical readings from the same patient (Patient ID), multiple transaction logs from the same user (User ID), or audio recordings from the same recording hardware.
- **Why It Matters (Data Leakage Prevention):**
 - If measurements from the same patient or user appear in both the training and validation sets, the model can exploit patient-specific idiosyncrasies rather than learning generalizable patterns, resulting in artificially inflated validation scores (data leakage).
 - Group-aware splitters (such as GroupKFold) enforce a strict boundary: **all samples sharing a given group ID are assigned together to either the training set or the validation set**, preventing any group overlap between train and test splits within the same iteration.

Shortcut Learning

occurs when models learn unintended decision rules that **perform well on standard benchmarks** but **fail to generalize** to real-world or out-of-distribution testing conditions

Key Drivers of Shortcuts:

- **Dataset bias (Shortcut opportunities):** Systematic co-occurrences in data allow networks to rely on superficial cues, such as predicting "cow" based on a grassy pasture background rather than animal shape. Scaling up dataset size alone does not eliminate these systematic biases.
- **Discriminative feature learning:** DNNs frequently exploit local textures or imperceptible high-frequency patterns over global shape, showing excessive invariance to essential semantic features once minimal discriminative signals are found.
- **Combined inductive biases:** Interactions between network architectures (e.g., lack of explicit shape priors), loss functions (e.g., cross-entropy stopping once separation is achieved), and optimizers (e.g., SGD favoring simple functions) naturally steer models toward shortcut solutions.

Shortcut Learning

Cases of the shortcut learning:

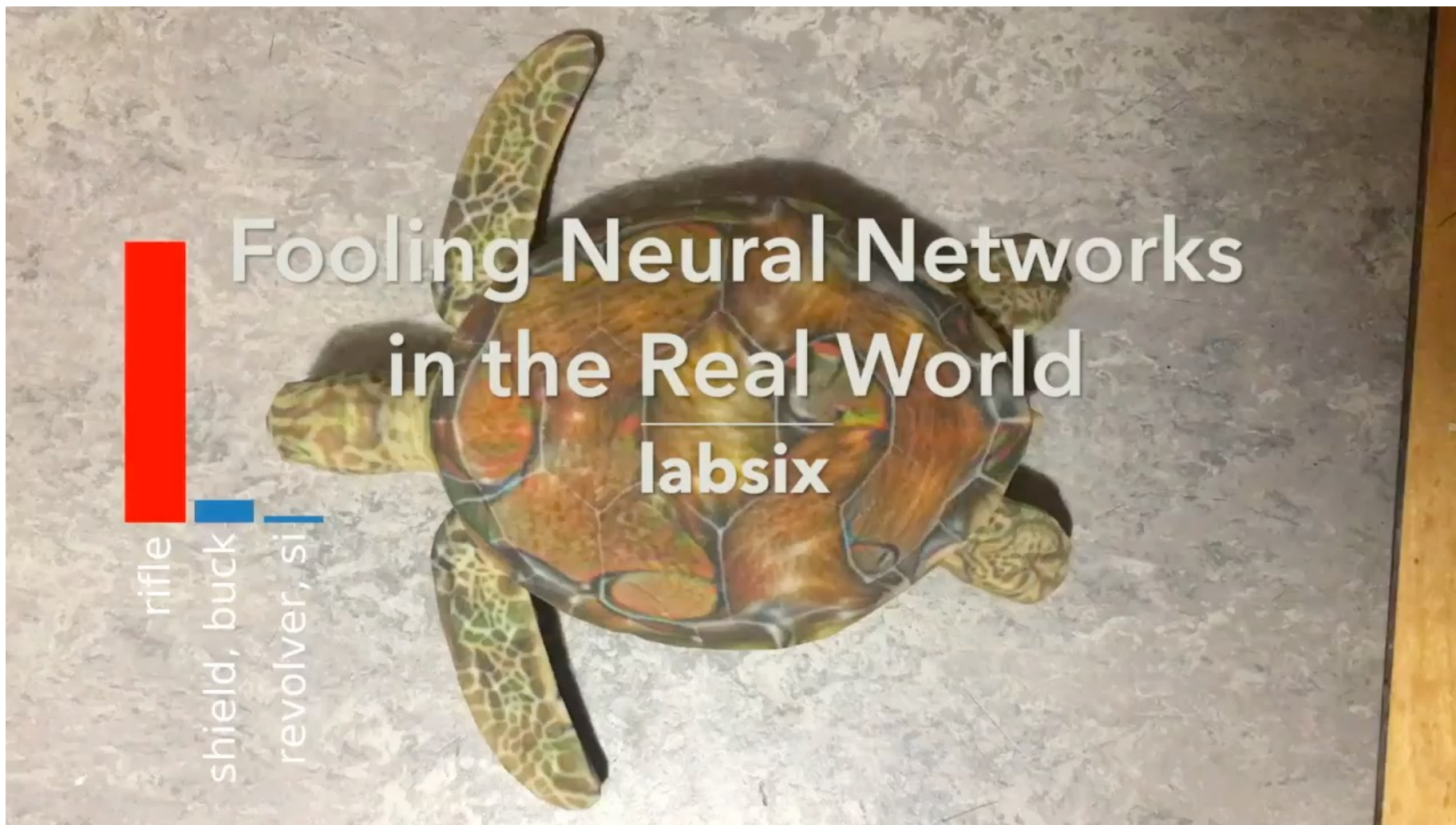
- **Medical Imaging:** Models detect pneumonia using hospital-specific metal identification tags rather than lung pathology, and standard classifiers are derailed by minor rotations, background changes, or adversarial perturbations.
- **Natural Language Processing:** Models like BERT exploit superficial heuristics such as the presence of negation tokens ("not") or sentence length rather than actual semantic argumentation.
 - ex) It **cannot** be **better**
You **cannot** achieve success **without** effort.
- **Reinforcement Learning:** Agents exploit reward function loopholes ("reward hacking"), such as permanently pausing Tetris to avoid losing.
 - ex) will be shown in next slide

Shortcut Learning



Computer program that learns to play classic NES games

Shortcut Learning

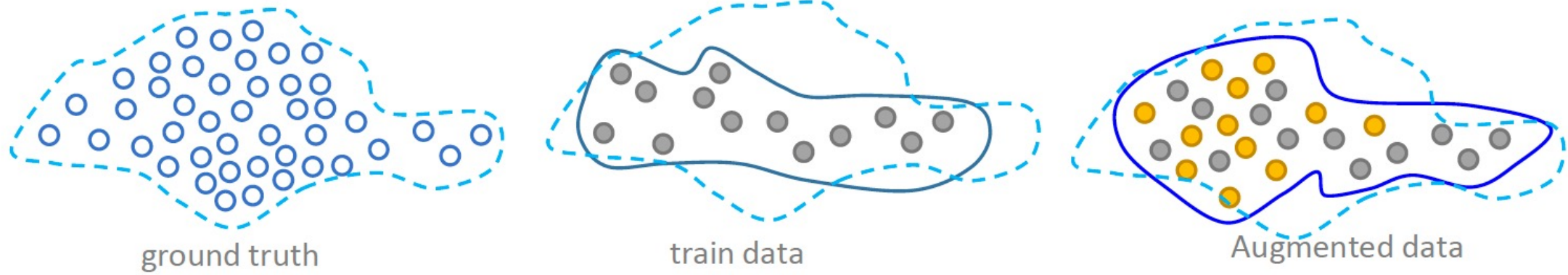


Solutions to the Problems?

In this lecture

- Integrating domain-specific priors and aggressive **data augmentation** strategies to eliminate spurious dependencies.
- Connecting **adversarial robustness** and **causal inference** to guide representations toward human-aligned, invariant mechanisms.
- Applying **domain generalization**, **domain randomization**, and **generative disentanglement** to isolate causal factors from environmental shifts.

Data Augmentation



Data Augmentation1 – Rule-based

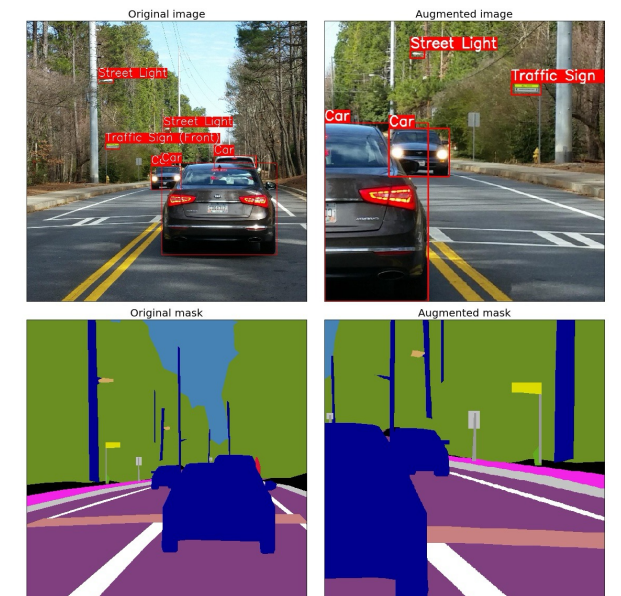
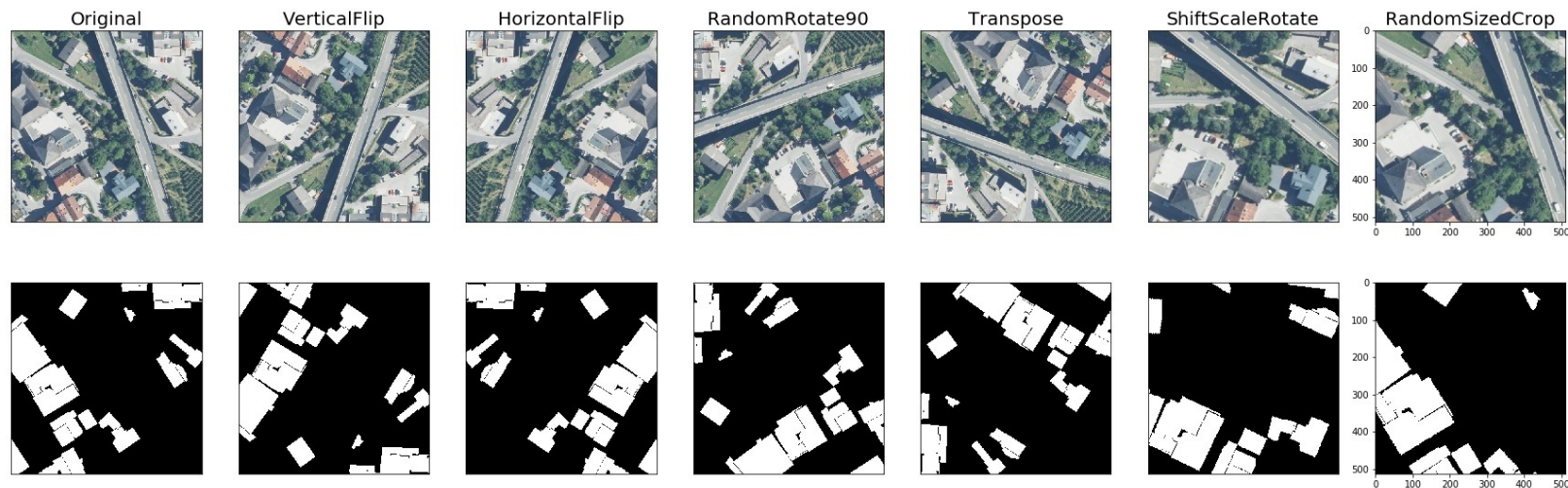
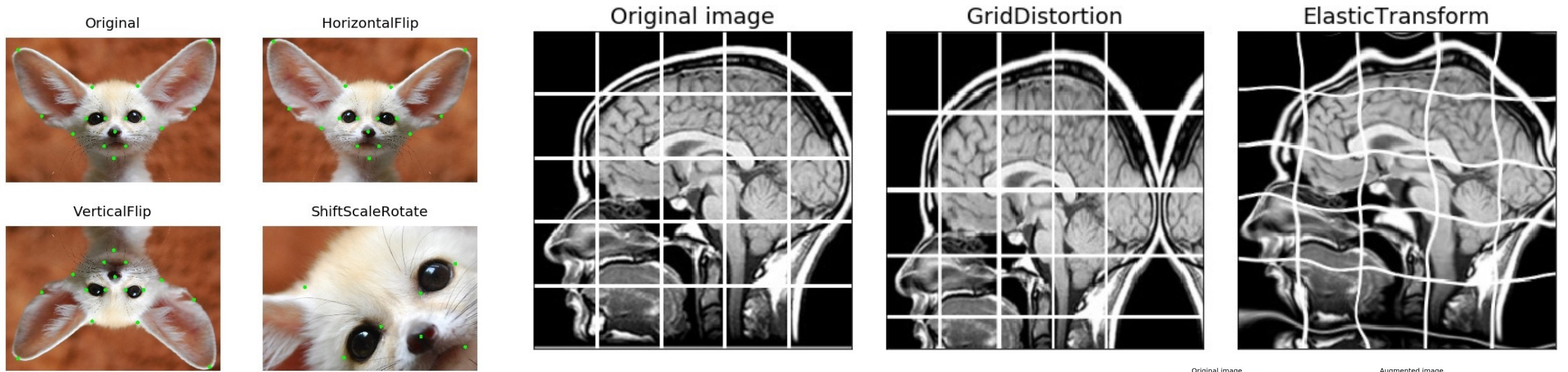


<https://github.com/albumentations-team/albumentations>

	Image	Heatmaps	Seg. Maps	Keypoints	Bounding Boxes, Polygons
<i>Original Input</i>					
Gauss. Noise + Contrast + Sharpen					
Affine					
Crop + Pad					
Fliplr + Perspective					

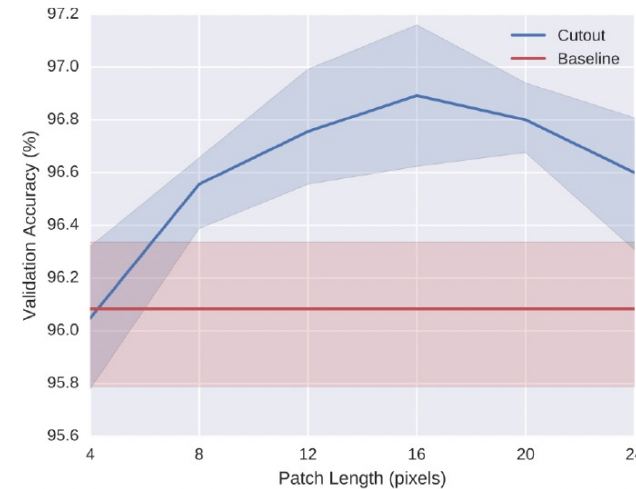
<https://github.com/aleju/imgaug>

Data Augmentation1 – Rule-based

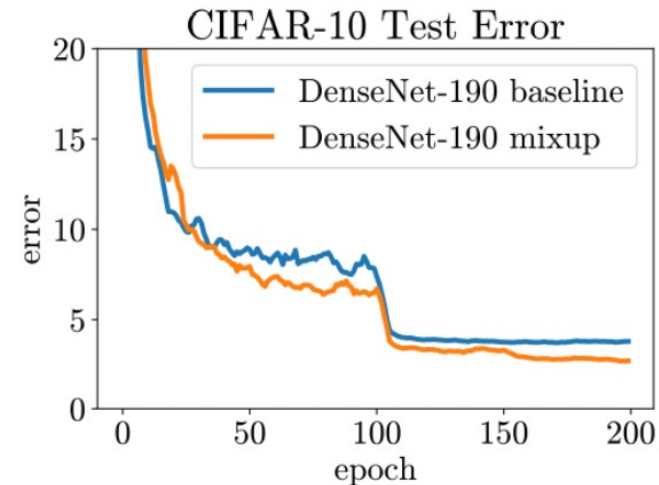
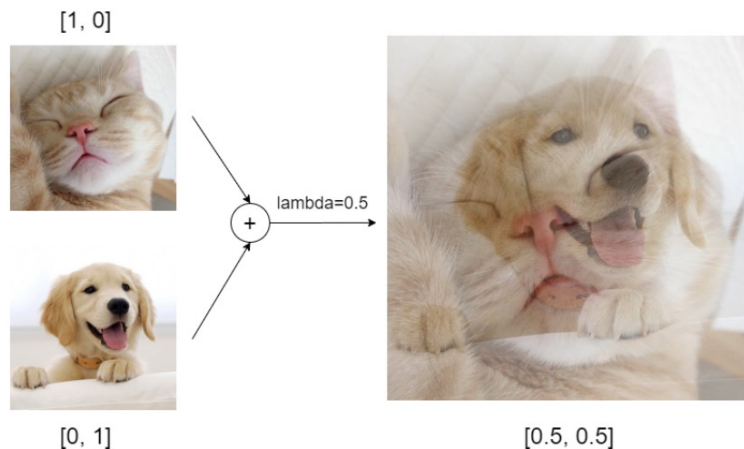


Data Augmentation1 – Rule-based

Cutout: Randomly replace square patches with noise.



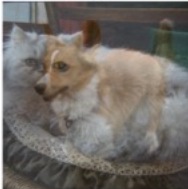
Mixup: Convex combination of pairs of images and their class labels.

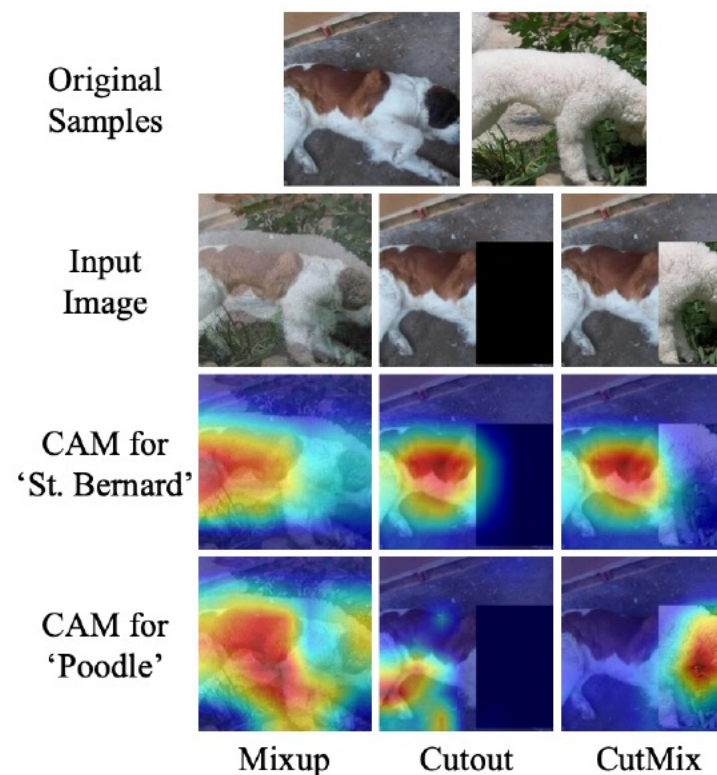


Data Augmentation1 – Rule-based

Cutmix:

- Patches are cut and pasted among training images.
- Class labels are also mixed proportionally to the area of the patches

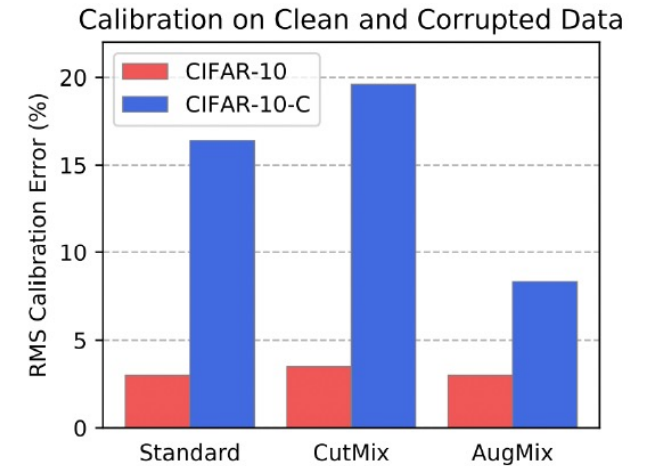
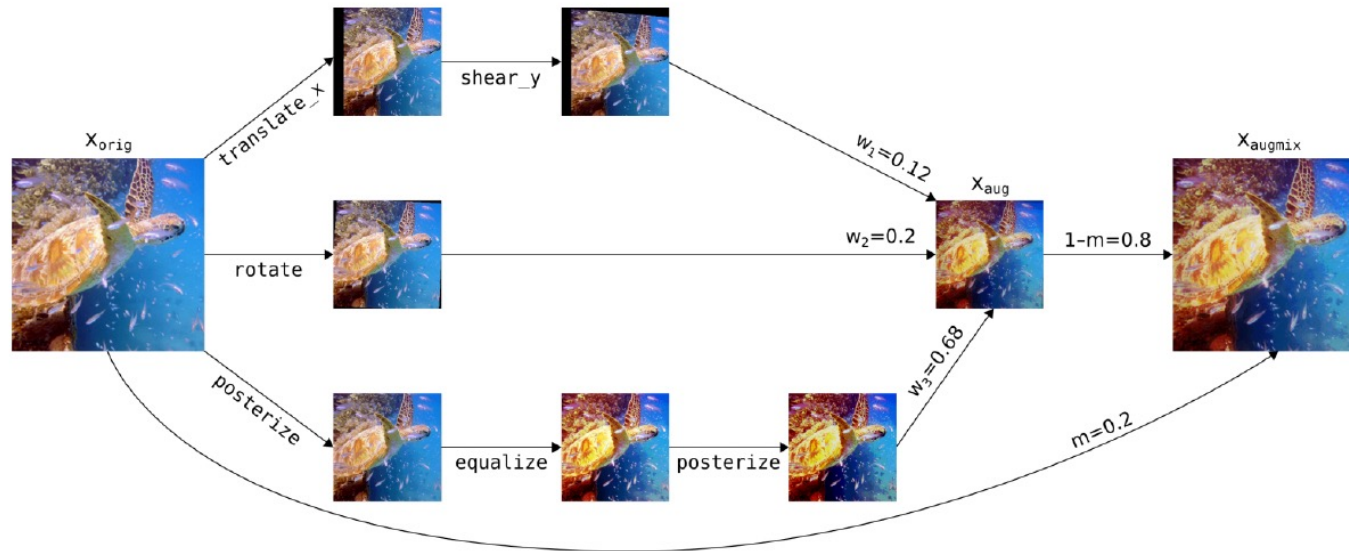
Image	ResNet-50	Mixup [48]	Cutout [3]	CutMix
				
Label	Dog 1.0	Dog 0.5 Cat 0.5	Dog 1.0	Dog 0.6 Cat 0.4
ImageNet Cls (%)	76.3 (+0.0)	77.4 (+1.1)	77.1 (+0.8)	78.6 (+2.3)
ImageNet Loc (%)	46.3 (+0.0)	45.8 (-0.5)	46.7 (+0.4)	47.3 (+1.0)
Pascal VOC Det (mAP)	75.6 (+0.0)	73.9 (-1.7)	75.1 (-0.5)	76.7 (+1.1)



Data Augmentation1 – Rule-based

Augmix:

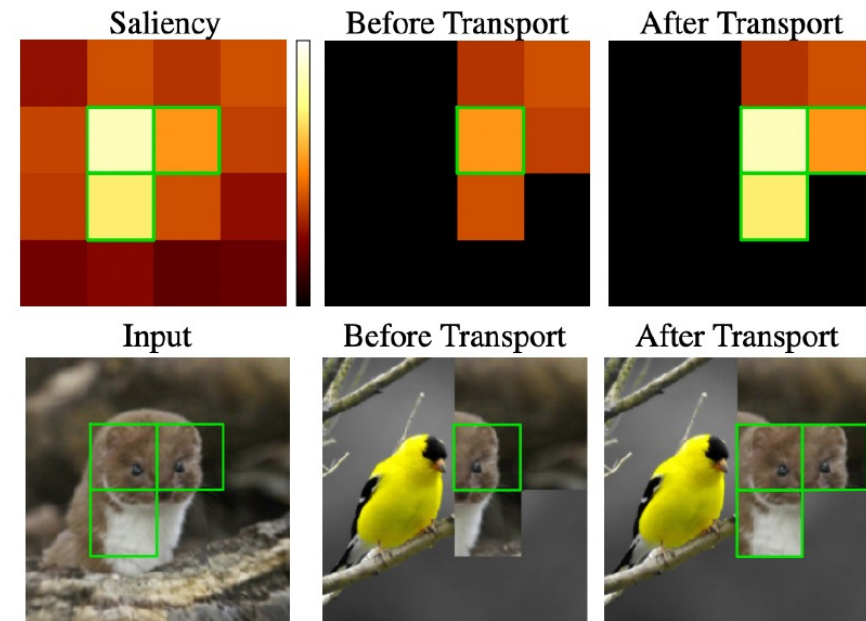
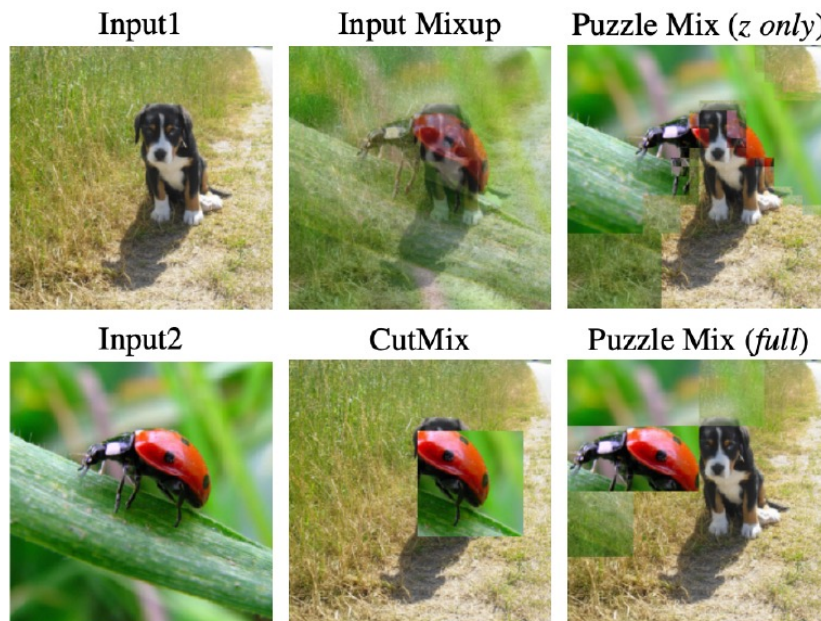
- Create multiple augmented images and mix them.
- Augmentation operations and the mixing weights are randomly sampled.
- Improves noise robustness and uncertainty estimates.



Data Augmentation1 – Rule-based

Puzzle Mix:

- Utilize the regional saliency information of natural images.
- Solve an binary transport problem to find the optimal move that maximizes saliency.



Data Augmentation1 – Rule-based

RandAugment (RA):

- Randomly sample a subset from a predefined set of 14 image transforms.
- Sequentially apply them with random distortion magnitudes.

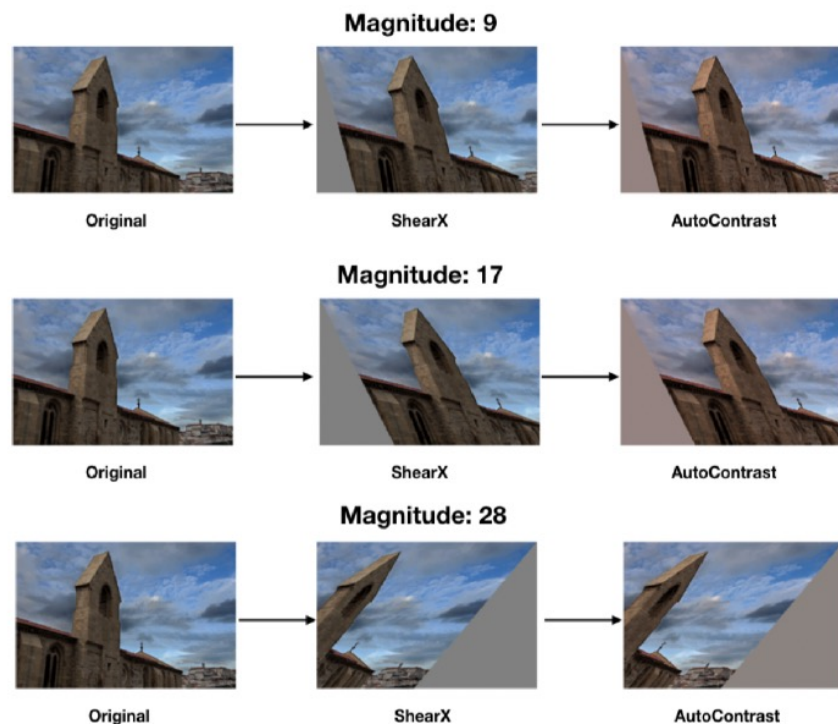
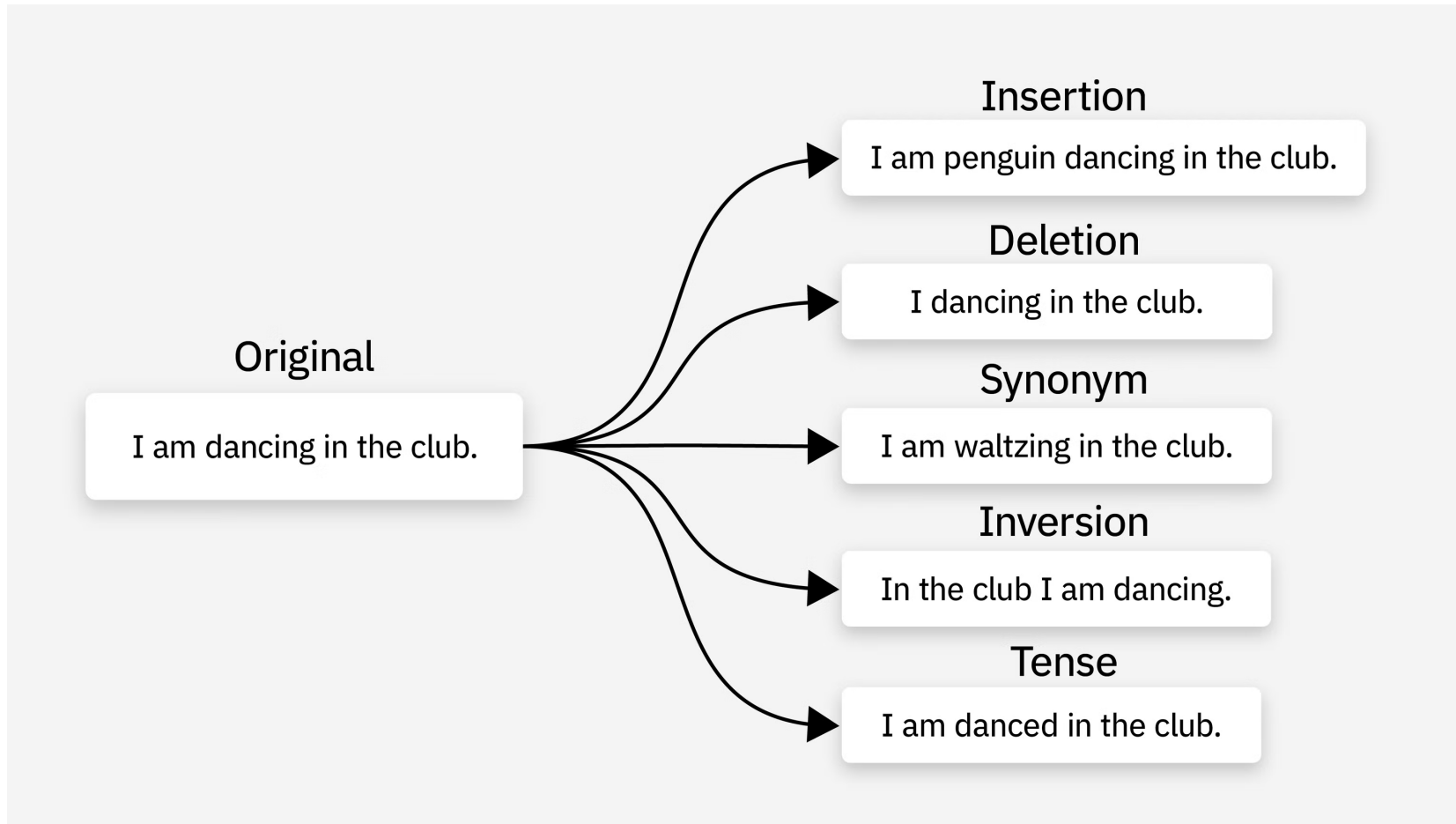


Figure 1. Example images augmented by RandAugment.

	baseline	PBA	Fast AA	AA	RA
CIFAR-10					
Wide-ResNet-28-2	94.9	-	-	95.9	95.8
Wide-ResNet-28-10	96.1	97.4	97.3	97.4	97.3
Shake-Shake	97.1	98.0	98.0	98.0	98.0
PyramidNet	97.3	98.5	98.3	98.5	98.5
CIFAR-100					
Wide-ResNet-28-2	75.4	-	-	78.5	78.3
Wide-ResNet-28-10	81.2	83.3	82.7	82.9	83.3
SVHN (core set)					
Wide-ResNet-28-2	96.7	-	-	98.0	98.3
Wide-ResNet-28-10	96.9	-	-	98.1	98.3
SVHN					
Wide-ResNet-28-2	98.2	-	-	98.7	98.7
Wide-ResNet-28-10	98.5	98.9	98.8	98.9	99.0

Table 2. Test accuracy (%) on CIFAR-10, CIFAR-100, SVHN and SVHN core set.

Data Augmentation1 – Rule-based



Data Augmentation1 – Rule-based

Thesaurus-based Substitution:

Replace a random word with its synonym using a Thesaurus.



Word Embedding-based Substitution:

- Use pre-trained word embeddings such as Word2Vec, GloVe, FastText, etc.
- Find the nearest neighbor words and substitute.

Nearest neighbors in word2vec



It is awesome → It is amazing
It is awesome → It is perfect
It is awesome → It is fantastic

Data Augmentation1 – Rule-based

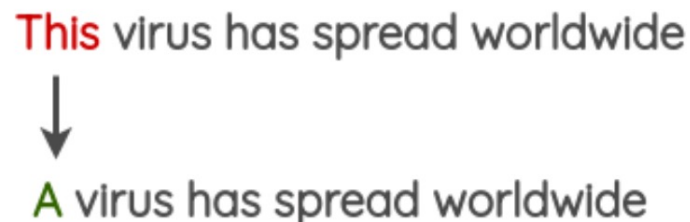
Masked Language Model:

- Use pre-trained masked language models such as BERT, ROBERTa, and ALBERT.
- Mask out some words & see what the model predicts.



TF-IDF-based Replacement:

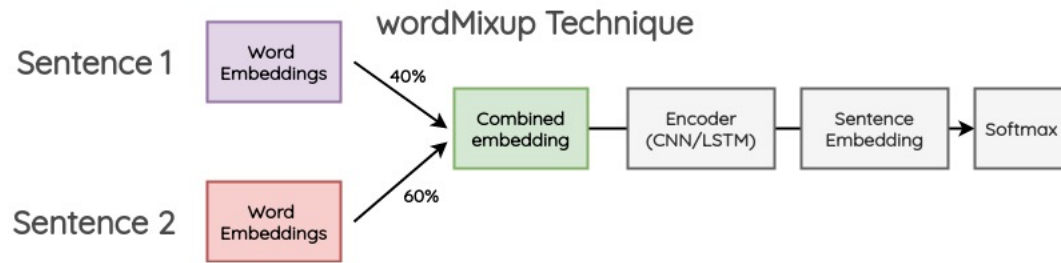
- Words with low TF-IDF scores are uninformative.
- Replacing those words will not affect the original semantic information.



Data Augmentation1 – Rule-based

Word Mixup:

- Use Mixup on word embedding features



Sentence A ($y_A = 1.0$, Positive):

"The movie was great"

→ Embedding sequence: $[E(\text{The}), E(\text{movie}), E(\text{was}), E(\text{great})]$

Sentence B ($y_B = 0.0$, Negative):

"That film was terrible"

→ Embedding sequence: $[E(\text{That}), E(\text{film}), E(\text{was}), E(\text{terrible})]$

Input (Novel Vector Sequence):

- Location 1: $0.7 \times E(\text{The}) + 0.3 \times E(\text{That})$
- Location 2: $0.7 \times E(\text{movie}) + 0.3 \times E(\text{film})$
- Location 3: $0.7 \times E(\text{was}) + 0.3 \times E(\text{was})$
- Location 4: $0.7 \times E(\text{great}) + 0.3 \times E(\text{terrible})$

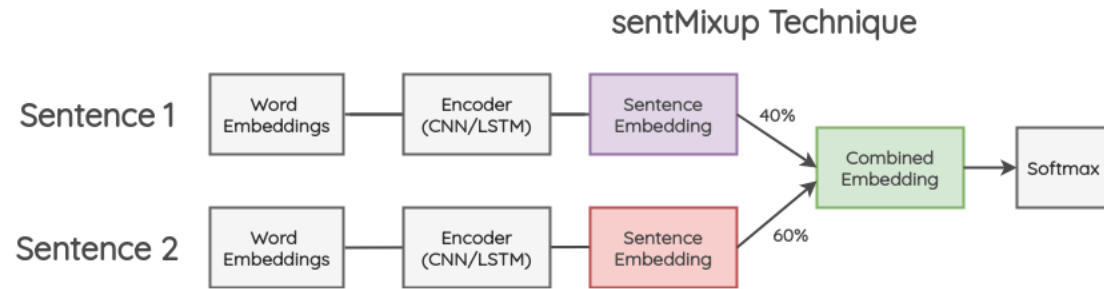
Novel Target Label:

- $\tilde{y} = 0.7 \times 1.0 + 0.3 \times 0.0 = 0.7$ (70% Positive)

Data Augmentation1 – Rule-based

Sent Mixup:

- Use Mixup on sentence embedding features



Sentence A (label: $[1,0]$):

- "I loved the acting and clever storyline." $\rightarrow v_A$ (label: $[0,1]$):
- "Completely boring and a total waste of time." $\rightarrow v_B$

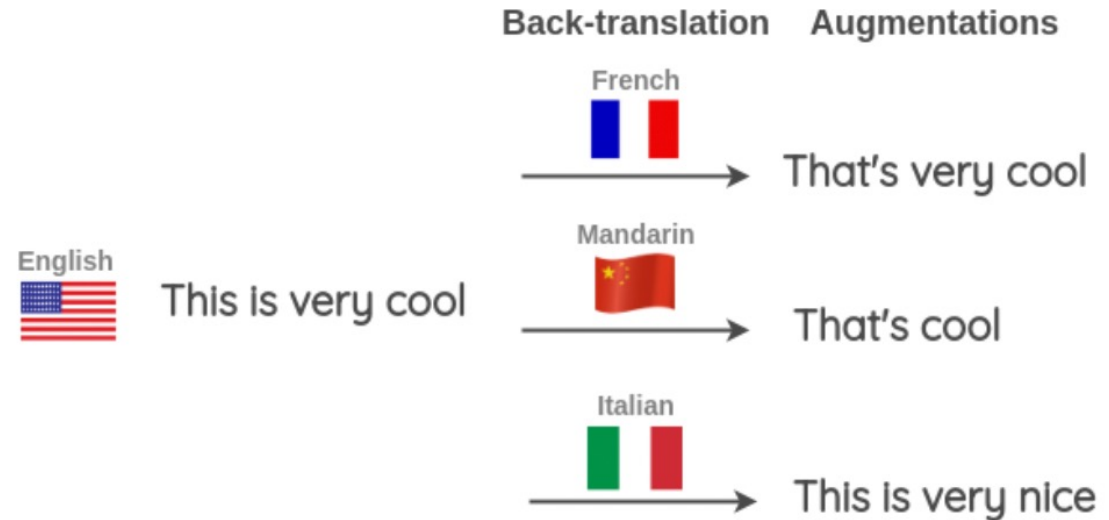
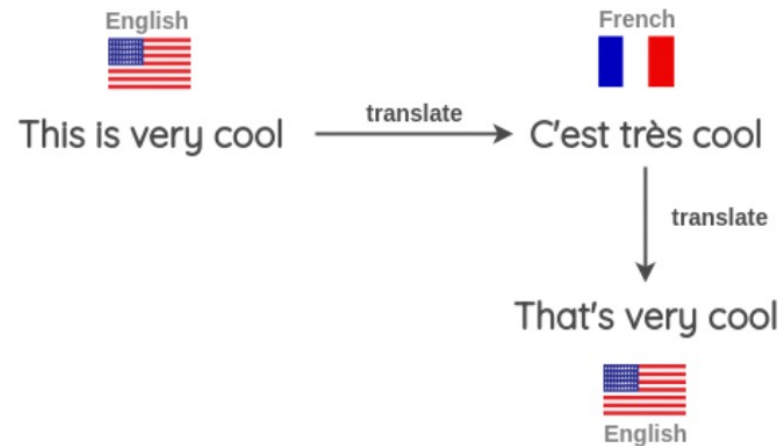
sentMixup result (neutral):

- **Input vector:** $\tilde{v} = 0.5 \times v_A + 0.5 \times v_B$
- **Target vector:** $\tilde{y} = 0.5 \times [1,0] + 0.5 \times [0,1] = [0.5,0.5]$

Data Augmentation1 – Rule-based

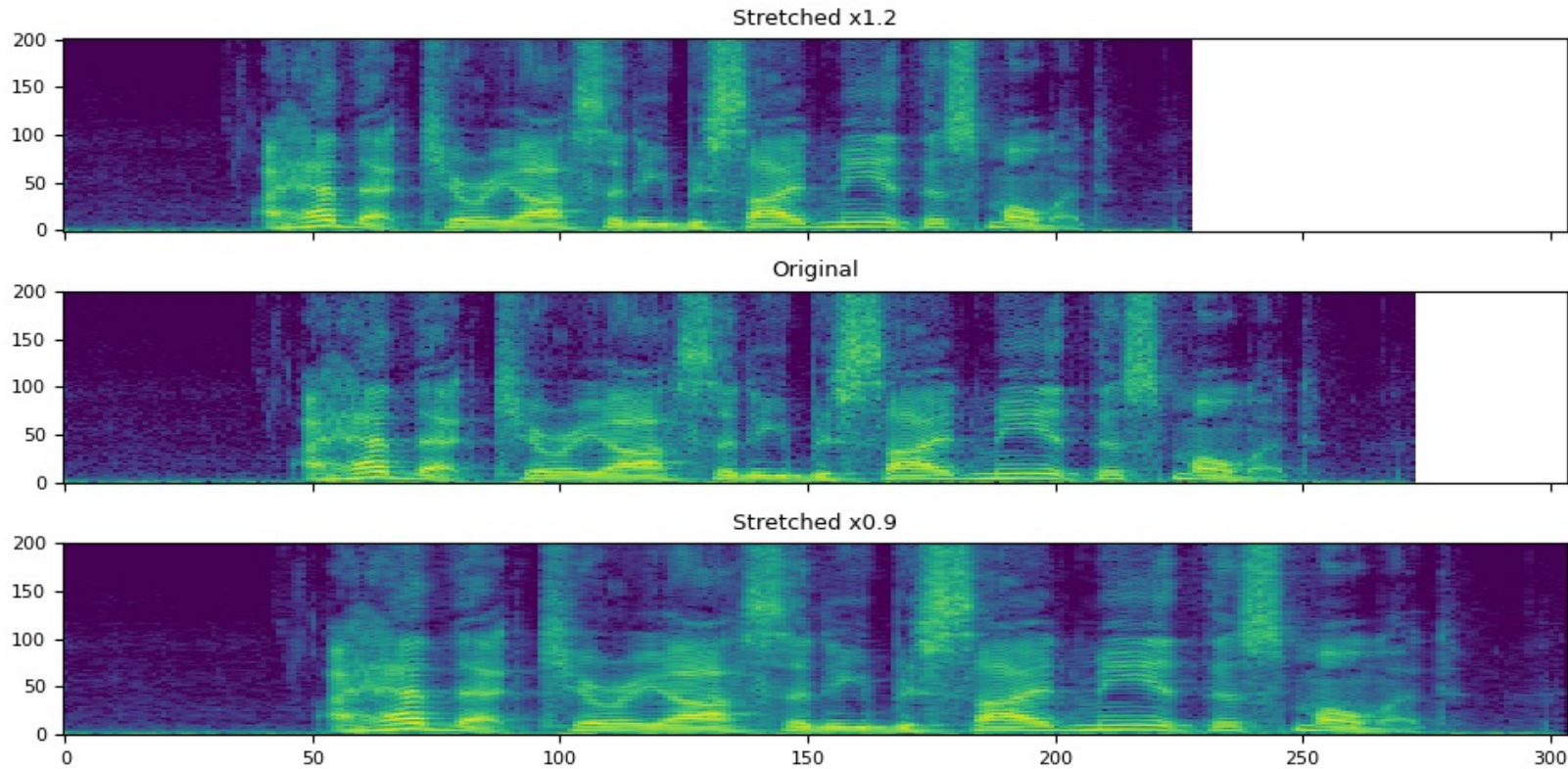
Back Translation:

- Translate into another language.
- Translate back to original language.



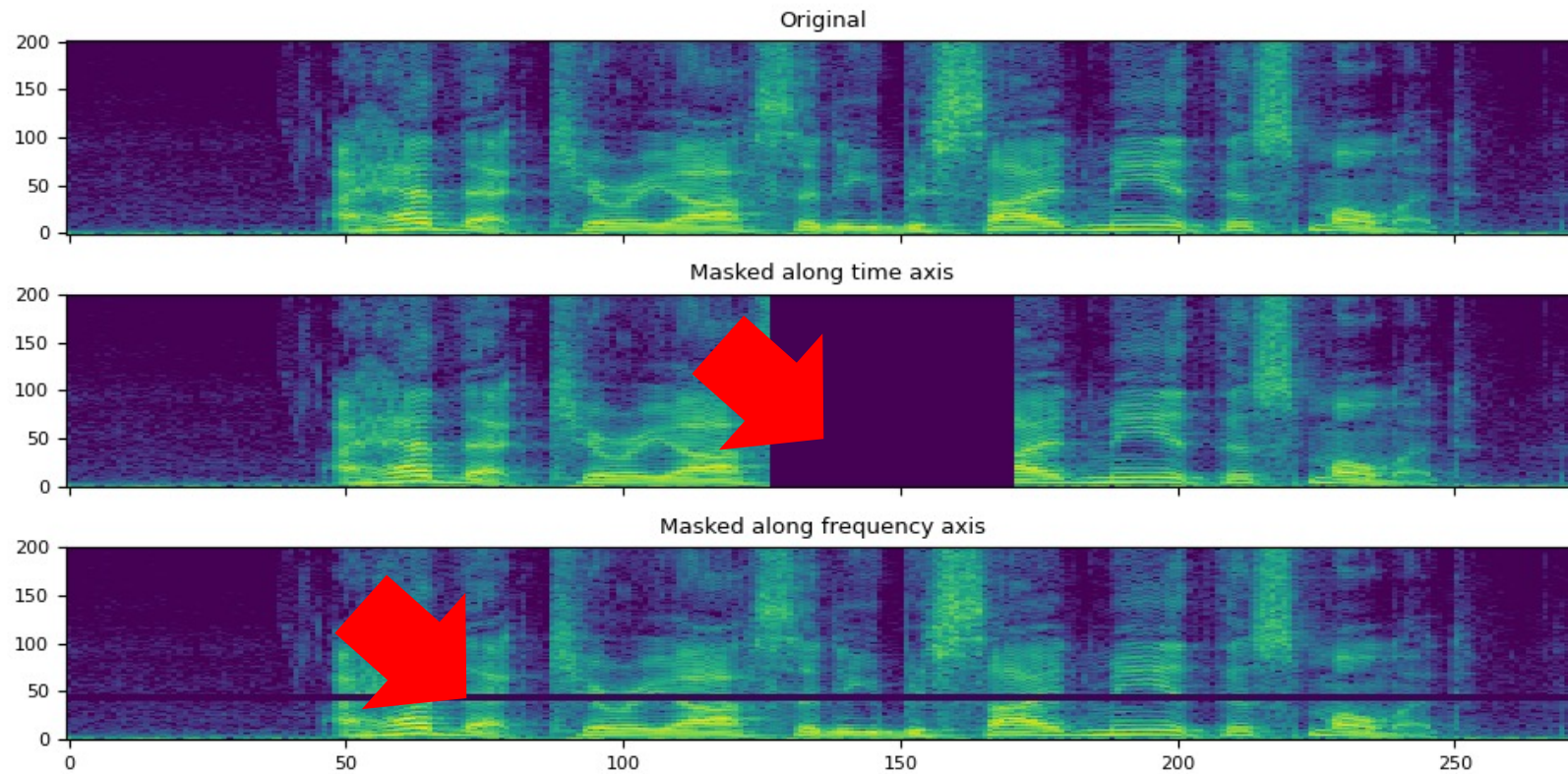
Data Augmentation1 – Rule-based

SpecAugment is a popular spectrogram augmentation technique.

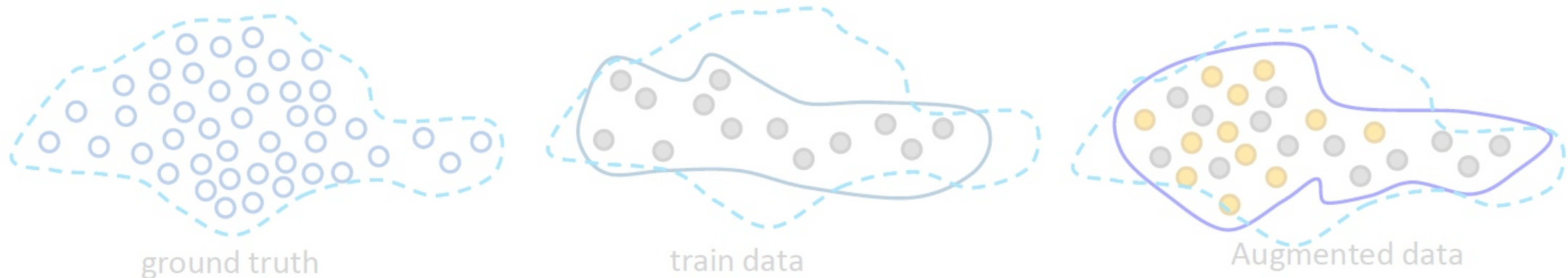


Data Augmentation1 – Rule-based

Time and Frequency Masking

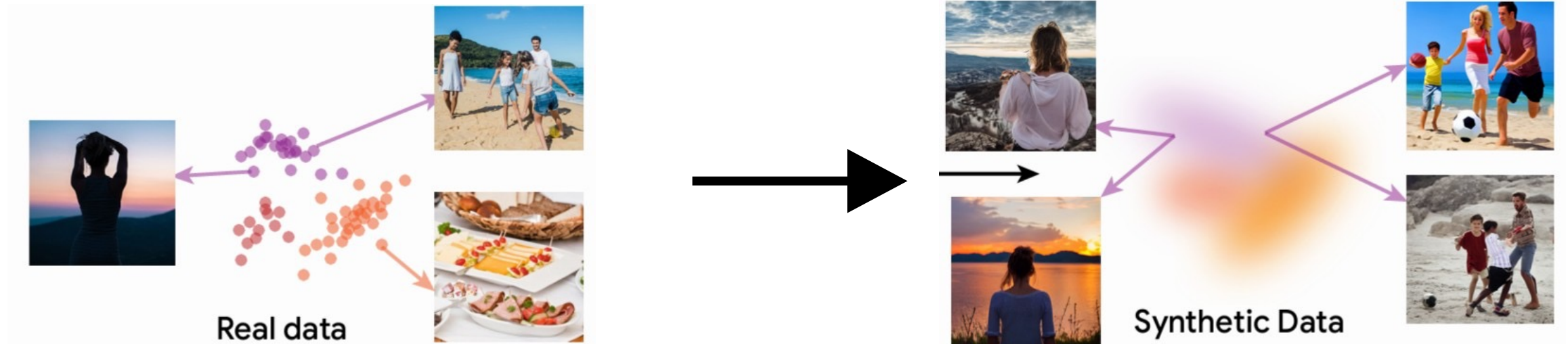


Data Augmentation2 – Data Generation

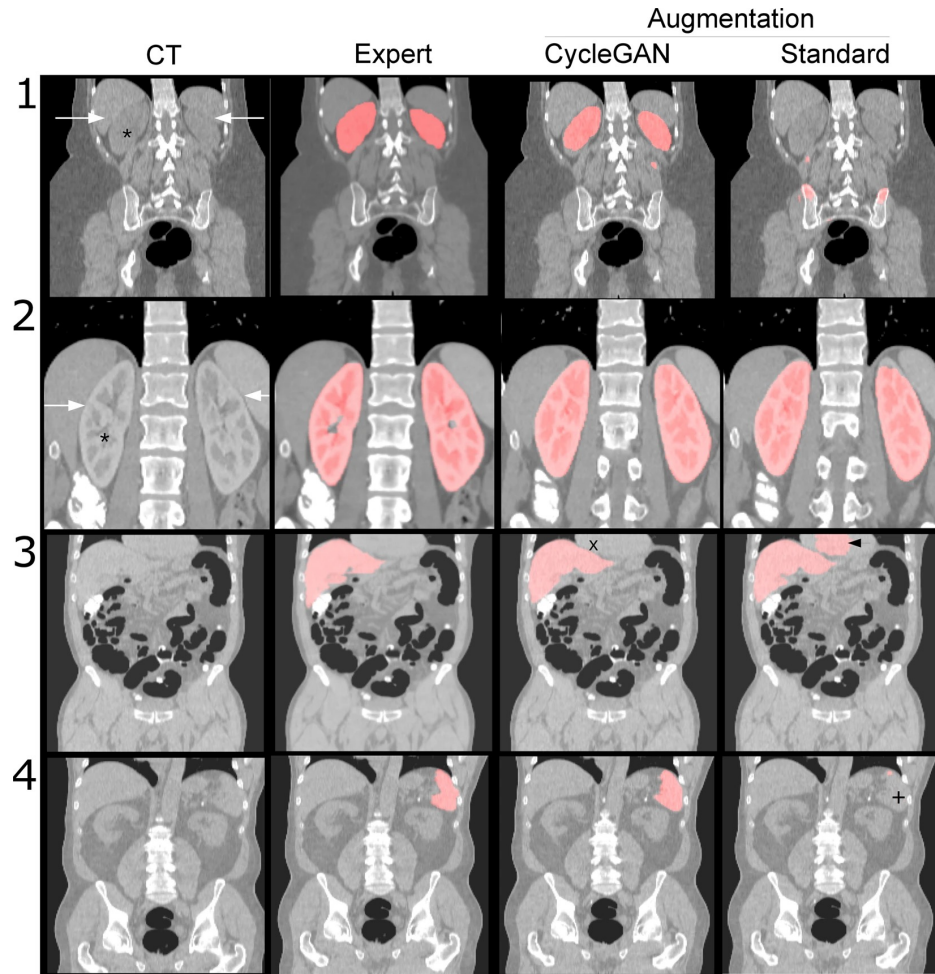


Small Data

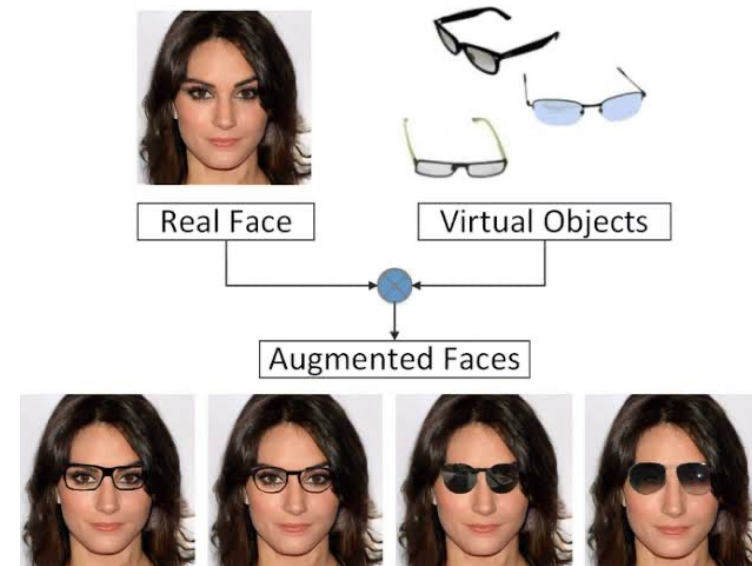
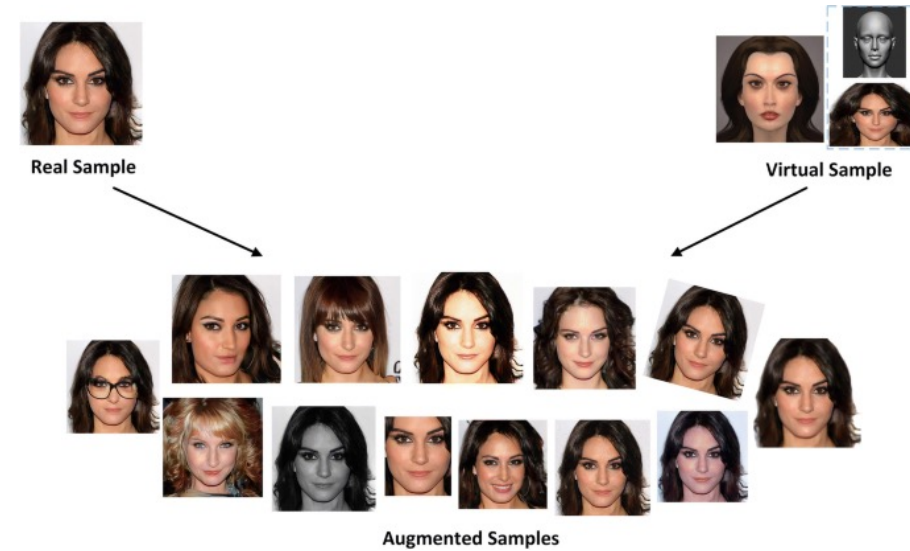
Data Distribution



Data Augmentation2 – Data Generation



Data augmentation using generative adversarial networks (CycleGAN) to improve generalizability in CT segmentation tasks



Data Augmentation2 – Data Generation

Conditional Pre-trained Language Models:

- Prepend the class label to each text in your training data
- Finetune a large pre-trained language model(BERT/GPT2/BART) on this modified training data. For GPT2, the fine-tuning task is generation while for BERT, the goal would be a masked token prediction.
- Using the fine-tuned language model, new samples can be generated by using the class label and a few initial words as the prompt for the model. This work uses 3 initial words of each training text and also generates one synthetic example for each point in the training data.

Original Training Data

Text	Label
It is very useful app	POSITIVE
The app was bad	NEGATIVE
...	...

→ POSITIVE<SEP>It is very useful app<EOS>

Finetune on training data



Task: Learn to generate training data

Output: POSITIVE<SEP>It is very useful app<EOS>

Data Augmentation3 –Synthetic Dataset

Playing for Benchmarks

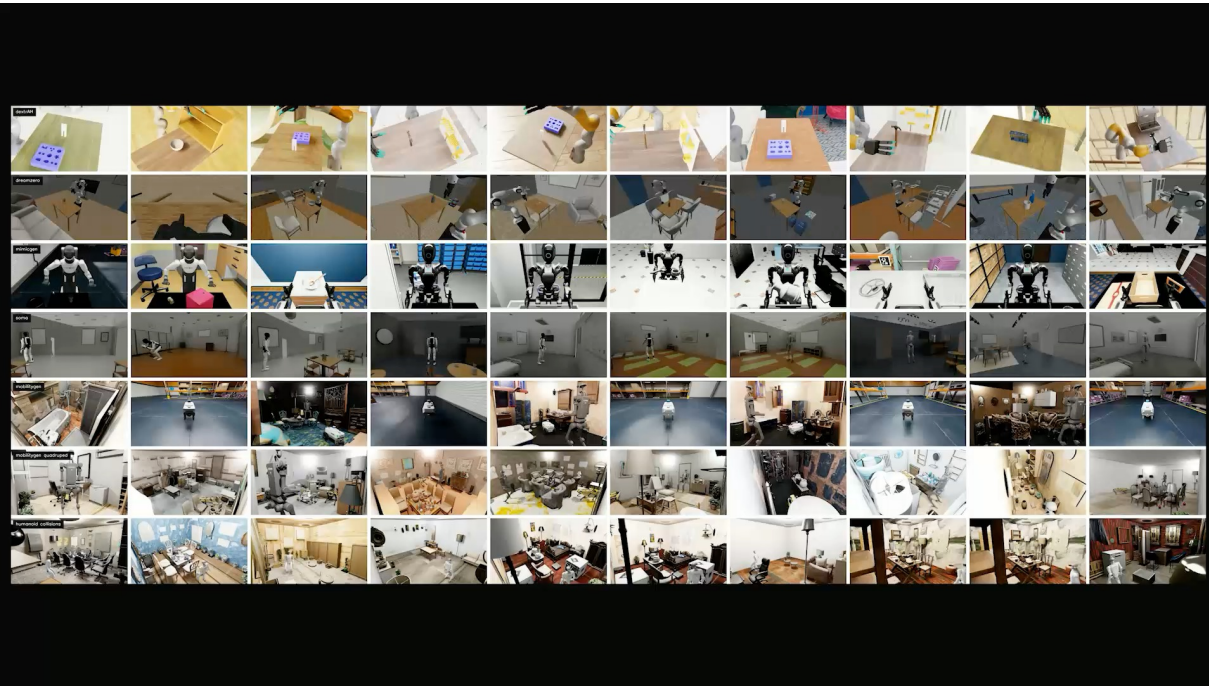
Stephan R. Richter
TU Darmstadt

Zeeshan Hayder
ANU

Vladlen Koltun
Intel Labs

ICCV 2017

Data Augmentation3 –Synthetic Dataset



Robot Simulation



World Model Simulation

Summary

1. Data Representation across Modalities

Text (NLP):

- ✓ Overcame limitations of Bag-of-Words (BoW) & TF-IDF.
- ✓ Solved Out-of-Vocabulary (OOV) challenges via subword tokenization (**BPE**, **WordPiece**).

Audio & Time-Series:

- ✓ Converted time-domain signals to frequency representations using Fourier Transforms (FT, STFT).
- ✓ Leveraged **Spectrograms**, Mel-Spectrograms, and MFCCs to process audio signals similarly to 2D image data.

2. Dataset Pitfalls & Generalization Challenges

Underfitting vs. Overfitting: Balancing model capacity and generalization error.

Data Leakage:

- ✓ Preventing test information from corrupting preprocessing, feature selection, or training pipelines.
- ✓ Applying robust evaluation splits (**StratifiedKFold**, **GroupKFold** for inter-sample dependencies).

Shortcut Learning:

- ✓ Mitigating models' tendency to rely on superficial correlations (e.g., backgrounds, textures, negation cues) rather than true causal semantics.

Summary

3. Data Augmentation Strategies

Rule-based Augmentation:

- ✓ **Vision:** Pixel/geometric transforms, **Cutout**, **Mixup**, **CutMix**, **AugMix**, **Puzzle Mix**, and **RandAugment**.
- ✓ **NLP:** Thesaurus/Embedding substitution, Masked Language Models (MLM), Back Translation, and Word/Sent Mixup.
- ✓ **Audio:** Time/Frequency stretching and masking (**SpecAugment**).

Generative & Synthetic Approaches:

- ✓ Generative models (e.g., CycleGAN, Conditional PLMs) to expand data distributions.
- ✓ High-fidelity synthetic datasets via game engines and robot/world model simulations.

Key Takeaway

"Effective deep learning requires principled modality-specific representations, strict evaluation setups to prevent data leakage, and aggressive data augmentation to overcome shortcut learning and ensure robust real-world generalization."